

Album Financiero: A Business Intelligence Solution for Scalable Personal Financial Analytics

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Tools Used:

Power BI, Microsoft Excel, PostgreSQL, SQL, Python, Power Query, DAX measures, DAX studio, Google Data Analytics Framework and Open AI (ChatGPT).

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Capstone Project – Business Intelligence & Analytics Portfolio

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Introductory note – document structure

This report is structured into three main sections, each addressing a specific stage of the project and its analytical intent:

Analysis Frameworks

This section defines the objectives, guiding principles, and key considerations of the project. It establishes what the analyst aims to achieve, the assumptions carried forward from Phase I, and the Business Intelligence perspective guiding Phase II.

Development Section

This section documents the execution of the project. It covers the practical work performed across the six phases of the Google Data Analytics framework, including data preparation, transformation, modeling, visualization, and validation.

Conclusions & Reflections

This section presents the final assessment of the project's outcomes. It includes lessons learned, reassessments made during execution, and reflections on data quality, automation, and the evolution of the analytical approach throughout Phase II.

Appendix and comments of the author

This section presents the location of documents informed between the report document, direct links to the Github repository and Google Site portfolio created, and the definition of colors used to differentiate the different columns, concepts, tables, and different archives and folders used over the project.

Analysis Framework

Introduction

This project is a continuation of the personal capstone “Album Financiero: A Full-Scale Personal Financial Analysis and Reporting Solution using Power BI,” referred to as Phase I. The objective of this phase is to evolve the existing solution into a more efficient and adaptable system by incorporating programming to automate data preparation processes, allowing Power BI to focus on visual analytics and reducing DAX complexity.

This phase, referred to as Phase II, focuses on optimizing the metrics and visuals derived from the “`album_financiero`” dataset by improving report organization, usability, and navigation. It also addresses performance improvements through the automated generation of curated datasets and their integration into PostgreSQL, enabling the creation of curated datasets to support more robust analytical modeling within the Power BI solution.

Project background

During the development of Phase I, several limitations were identified in the way data preparation and transformation were handled within Power BI. As the volume of information and the number of required metrics increased, it became evident that relying solely on Power BI and DAX for the full data preparation workflow introduced unnecessary complexity and reduced flexibility.

Through an iterative review of the datasets generated in Phase I, and with the support of OpenAI as a design and validation tool, opportunities were identified to restructure the data preparation process using SQL and Python. These technologies were selected to automate transformations, standardize outputs, and generate curated, Power BI-ready datasets that could be imported directly into the model.

By shifting a significant portion of the data processing logic to SQL and Python, Phase II allows Power BI to focus on its core strength: visual analytics and metric consumption. This redesign reduces DAX complexity, improves model maintainability, and enables the creation of more direct, scalable, and metric-focused measures.

A clear problem statement / research question

Building on the successful foundation established in Phase I, the primary goal of this project is to further optimize the Album Financiero solution by improving

efficiency, scalability, and maintainability through the integration of Excel optimization, SQL, and Python.

This objective is guided by the following questions:

- How can the existing Album Financiero workflow be enhanced through automation and optimized data preparation processes to reduce manual effort while preserving the accuracy and reliability of financial insights?
- How can the underlying Excel-based data sources be restructured and standardized to better support scalable analytics, consistent metrics, and long-term reporting in Power BI?
- How can a functioning and validated Business Intelligence solution be evolved into a more efficient and automated system that improves performance, reduces operational complexity, and supports future analytical growth?

Project core principles

The project is guided by a set of core principles aimed at optimizing the Album Financiero workflow established in Phase I while strengthening Business Intelligence best practices.

The Power BI solution is designed to be as efficient as possible, limiting its responsibility to managing essential relationships and visual interactions. Data preparation, transformation, and standardization are handled upstream to minimize DAX complexity and ensure that measures remain direct, readable, and focused on metric definition rather than data manipulation.

Source Excel documents used by the user are transformed into standardized CSV data sources and imported into PostgreSQL using DBeaver. This database layer is used to generate curated datasets specifically designed to feed the Power BI model, ensuring consistency, scalability, and alignment with a star schema-based structure. These curated datasets are primarily technical assets and serve as the single source of truth for the Power BI solution.

Python is used to support automation of dataset generation and update processes, enabling periodic refreshes with minimal manual intervention. This approach allows the user to focus on updating information and consuming insights within Power BI, rather than managing repetitive data preparation tasks.

In addition to technical optimization, this project emphasizes continuous skill development. Phase II intentionally incorporates tools, workflows, and design logic commonly used by data analysts working in Business Intelligence environments. OpenAI is used as a support tool throughout the project to assist with debugging,

validation, and refinement of code and logic. Its role is to support understanding and decision-making, not to replace analytical reasoning or ownership of the solution.

Project Objectives

1. Analyze existing DAX and calculated columns: Understand the current logic in Power BI, evaluate which measures support the visuals, and identify opportunities to simplify or pre-calculate metrics outside of Power BI.
2. Define SQL and Python responsibilities: Decide which calculations, data transformations and automatization processes should be handled with SQL statements or Python codes to improve efficiency, automation, and scalability, while keeping essential DAX measures to Power BI.
3. Rebuild and prepare processed data sources: Apply the improvements identified in Phase I by adding the required IDs and structural columns using Power Query and pivot tables, removing redundant or derivable fields, and generating clean, standardized CSV versions of each spreadsheet. These processed CSV files become the primary data sources for analysis. The original Excel content is reorganized by separating tables that feed transactional information (fact tables) from supporting tables that provide contextual or explanatory information (reference and dimension tables). This reorganized set of Excel files is referred to as the Master Excel Documents.
4. Recreate and enhance visuals: Validate that all curated datasets created in PostgreSQL feed the core metrics correctly in Power BI, so we can rebuild the existing pages into more focused in details reports, with a better distribution of visuals and usage of more appealing information shown.
5. Optimize and automate refresh: Implement a scalable, low-maintenance workflow by ensuring that the system can be easily updated with new data over time without any big trouble to the final user.

Project scope and limitations

This project focuses on improving and automating the personal financial record maintained in the Album Financiero over the past 10 years. The scope includes:

- Master Excel document: The “**album_financiero**” Excel file will be cleaned, normalized and adapted with the help of Power Query, and will remain as the master document for ongoing updates; processed datasets are intended solely for analysis and automation purposes.

- Data cleaning, normalization, semi automatization of processes, and restructuring of core spreadsheets.
- Creation of processed and curated datasets to serve as the primary source for analysis.
- Integration of SQL and Python for preprocessing, automation, and enhanced efficiency.
- Development of Power BI reports and dashboards for descriptive and diagnostic analysis, including improved visuals, budgeting alignment, and pending payments tracking.

Limitations:

- Programming expertise: SQL and Python are applied using the knowledge acquired through prior training and coursework before the start of the project. While the implementation aims to achieve optimal and efficient results, alternative or more advanced coding approaches may exist that could further optimize performance or simplify certain processes.
- Analytical focus: The project remains centered on historical personal financial data. Predictive modeling, forecasting, and integration of external data sources are intentionally excluded to preserve the project's descriptive and diagnostic BI focus.
- Tool integration limitations: While automation will be applied, some processes may still involve manual review to ensure data quality and consistency.

Project timeline consideration

As in Phase I, this project is part of the analyst's professional development in data analysis and visualization, and therefore it is conducted without a fixed deadline. This approach allows sufficient time to explore advanced tools, deepen the understanding of Power Query, SQL, and Python, and apply best practices in data modeling, transformation, and dashboard design. From a Business Intelligence perspective, this flexibility supports iterative refinement of metrics, visuals, and data structures to ensure the solution delivers clear, reliable, and decision-oriented insights.

Data preparation and cleaning

Phase II focuses on strengthening the analytical foundation established in Phase I by ensuring that all data sources are structured, consistent, and optimized for automated Business Intelligence reporting. Rather than changing the financial logic of the original model, this phase enhances data usability and scalability to support

efficient metric calculation and reliable visualization in Power BI. Key activities include:

- **Data model alignment and relational readiness:** Introducing standardized ID fields (such as *id_deudas* and *id_presupuesto*) to formally link tables and enable stable, scalable relationships, while preserving the original business logic of the master document.
- **Standardization for analytical consumption:** Applying controlled transformations using SQL, Python, and Power Query to harmonize descriptions, categories, and labels, ensuring compatibility with a relational BI model and reducing downstream complexity in DAX measures.
- **Data integrity and analytical reliability:** Validating that all curated datasets remain fully consistent with the master Excel source and privacy requirements, guaranteeing that insights produced in Power BI accurately reflect the underlying financial records.

Environment & tools setup

- Microsoft Excel – Preparation of source data, use of Power Query, PivotTables, and formulas/functions for preliminary analysis and reference/dimensional table creation.
- Power Query – ETL layer used for data cleaning, transformation, and standardization prior to database ingestion and Power BI modeling.
- OpenAI (ChatGPT) – Support for coding assistance (Python, SQL, DAX), formula optimization, translations, and structured documentation drafting.
- PostgreSQL (via DBeaver) – Creation of the relational database, execution of SQL scripts, and generation of curated datasets.
- Python (Jupyter Notebook) – Automation of SQL execution and dataset generation for consumption by the Power BI solution.
- Anaconda Prompt – Creation of an executable (EXE) to automate the data pipeline with minimal user interaction.
- Power BI – Connection to curated datasets, semantic modeling, and development of analytical visuals supported by DAX measures.
- DAX Studio – Extraction and documentation of DAX measures and validation of table relationships within the Power BI model.
- Local execution environment (Windows OS) – Management of file paths, credentials, and scheduled or manual pipeline execution.

Development Procedure

This project showcases core Business Intelligence competencies, including data preparation, dimensional modeling, semantic layer design, and dashboard delivery. In Phase II, the original Excel and Power BI solution evolved into a scalable BI workflow by introducing a structured data layer and automated processing. Python is used to orchestrate ETL tasks, SQL to design and maintain curated and relational datasets, and Power BI to consume these datasets through optimized DAX measures and interactive reports. This architecture enables Power BI to focus on analytics and visualization, while data preparation and transformation are handled upstream.

The project follows the Google Data Analytics framework, selected for its clear alignment with real-world BI workflows and role responsibilities. This methodology structures the development process into well-defined phases that reflect how analytical solutions are designed, implemented, and communicated in professional BI environments:

- Ask: Define the problem and identify key questions to answer.
- Prepare: Collect and organize the data for analysis.
- Process: Clean and transform the data to ensure quality and consistency.
- Analyze: Explore the data, identify patterns, and develop insights.
- Share: Communicate findings clearly through visualizations and storytelling.
- Act: Apply insights to make informed decisions or recommendations.

Ask

Stakeholder: Juan Carlos Larios Coronado (also the analyst)

Objective: The objective of Phase II is to enhance the existing Album Financiero Business Intelligence solution by introducing a structured, automated, and scalable data workflow that improves analytical efficiency while preserving the usability achieved in Phase I. This phase focuses on strengthening the data foundation, optimizing metric calculation, and enabling Power BI to operate primarily as a reporting and insight-generation layer.

- Adapting the original “album_financiero” into a master document sets of documents to meet new technical requirements, while preserving its user-friendly structure. Leave the fact tables together in a single document, while the reference/dimensional tables will be separated into new documents each.
- Creating SQL-transformed tables that manage preprocessing, IDs, and relational structures, delivering the curated datasets that will feed the Power BI solution.
- Developing Python scripts to generate curated datasets and mainly to automate ETL processes, saving time, reducing manual work, and minimizing human error.
- Generating final curated datasets that are clean, normalized, and ready for analysis.
- Enhance the Power BI solution with new DAX measures, improved visuals, and optimizations based on the processed datasets, resulting in better performance and analytical depth.
- Applying data validation and quality controls across the pipeline to ensure analytical accuracy, reliability, and consistency.
- Producing clear and structured documentation to support transparency, reproducibility, and professional presentation of the end-to-end BI solution.

Project Roadmap: Key Steps to Follow

- Review existing DAX measures and calculated columns to understand the semantic model and analytical logic established in Phase I.
- Define clear responsibilities between SQL, Python, and Power BI, shifting data preparation and automation upstream to reduce complexity in DAX.
- Rebuild and prepare processed data sources by cleaning, normalizing, and adapting and generating reference/dimension tables from fact tables.
- Recreate and enhance dashboards by validating metrics, refining visuals, and ensuring insights remain consistent with business objectives.
- Optimize and automate data refresh processes to ensure scalability, reliability, and ongoing usability of the BI solution.

Prepare

The documentation required for this project builds upon the materials created in Phase I. These have undergone amendments and adaptations to fulfill the goals of this project.

Prepare – Master Excel Document “album_financiero”

The data source consists of more than ten years of personal finance records manually recorded in Excel and is distributed between five fact tables, a definition sheet, and two reference tables:

1. Each row in “**c_diaria**” represents a transaction, serving as the project’s fact table.
2. The “**presupuesto**” contains monthly and annual budgets established by the user to maintain control of personal finances.
3. The “**credito**” tab documents both settled and outstanding credit debts.
4. The “**viajes**” tab provides a historical log of trips taken over the years, including travel dates and brief descriptions or reasons for each trip.
5. The “**inversiones**” tab details the investments that have existed in the life of the “Album Financiero” spreadsheet.
6. The “**definiciones**” tab consolidates definitions and recommendations for the columns “*razon_uso*,” “*clave*,” and “*metodo_pago*.”
7. The “**parametros**” tab informs the total available amount for each managed credit account.
8. The “**geografia**” tab store complete geographical information for each place visited and recorder in the “**viajes**” tab.

As noted earlier in this report, the master Excel document was adjusted to support a more structured and scalable BI data model. These changes preserved the original financial logic while reorganizing the data sources to better separate transactional information from descriptive attributes.

Specifically, fact tables were retained within the main master document, while reference and dimension tables were extracted into dedicated spreadsheets:

- “**viajes**” became into the reference table “**tabla_id_viajes**”.
- “**geografia**” became into the reference table “**tabla_geografia**”
 - In “**tabla_geografia**” the columna “*moneda*” was created.

Prepare – “dax_mediciones_y_visuales”

The document “dax_mediciones_y_visuales” was created during Phase I of this project to document the relationship between Power BI visuals and their underlying DAX logic. This document contained the following columns:

- *nombre_visual*: Lists the name of each visual created in Power BI.
- *dax_medicion*: Identifies the corresponding measure(s) used in each visual. If a visual does not require a measure, the entry is “sin formula”.
- *dax_formula*: Provides the full DAX formula for each measure. If a visual does not require a measure, the entry is “sin formula”.
- *usado_denuevo*: Indicates whether the referenced measure is used more than once.

Initially, the documentation did not specify whether DAX measures were reused across visuals or whether calculated columns had been created within Power BI. These elements were later incorporated into the document, as they were considered essential for a complete understanding of the semantic layer and metric reuse within the solution.

Additional columns were also introduced to help identify frequently used measures and evaluate opportunities to replace complex DAX logic with upstream transformations in SQL or Python. The criteria and decisions related to this optimization are explained in detail in the corresponding section of the **Process phase**.

Process

During this phase, the full ETL workflow was designed and implemented. Existing data sources were restructured, new reference and dimension tables were created, and all supporting documents required to generate the curated datasets were produced. These curated datasets were then connected to the Power BI solution, where table relationships were established following a star schema design to support efficient analysis and reporting.

Original Approach for the Project

The initial approach for Phase II was to build upon the analytical logic developed in Phase I by reviewing the existing DAX measures and determining how SQL and Python could be incorporated to support data preparation and ETL activities. The objective was to simplify the semantic layer by shifting part of the transformation logic upstream, allowing Power BI to focus primarily on visualization and metric calculation.

Under this approach, the recommendations identified in Phase I were intended to be applied incrementally to the Excel source documents, ensuring compatibility with a more efficient Power BI solution. The optimization strategy prioritized DAX measures first, with SQL and Python introduced progressively as supporting tools for automation and preprocessing.

The planned steps were defined as follows:

- Step 1: Reanalyze all DAX measures to determine which calculations would remain in Power BI and which could be adapted or migrated to SQL or Python.
- Step 2: Define a clear division of responsibilities between SQL, Python, and Power BI for data processing and transformation.
- Step 3: Restructure the Excel data sources into normalized tables and automate dataset creation.

Actual approach for the project

As the project progressed, it became evident that the original execution plan did not fully align with the project's objectives or the structural reality of the data. While the initial analysis of DAX measures was performed as planned, it was determined that meaningful optimization could not be achieved without first addressing the underlying structure of the Excel data sources.

To implement the recommendations identified in Phase I, the master Excel set of documents required normalization and structural adjustments. These changes

directly affected existing DAX measures and Power BI visuals, making it impractical to optimize calculations in isolation. As a result, the initial strategy of reviewing and refactoring DAX measures individually was found to provide limited value in terms of efficiency and architectural clarity.

To better align the workflow with Business Intelligence best practices, a revised approach was adopted, shifting the focus from incremental DAX optimization to data model-driven preparation and engineering. The updated methodology emphasized building a robust and scalable data foundation before refining the semantic layer.

The revised steps were defined as follows:

- Step 1: Perform a partial review of the DAX measures created in Phase I, classifying them by complexity based on the number of nested or dependent measures. Calculations with higher complexity were identified as potential candidates for upstream optimization.
- Step 2: Adapt the original “**album_financiero**” Excel source into “**master_album_financiero**”, and create the necessary reference and dimension tables to support new identifiers, following the recommendations established in Phase I and Phase II.
- **Step 3:** Conduct a detailed review of DAX measures identified as candidates for migration, evaluating each calculation individually. A classification table “**datasets_generados**” was created to document and map the curated datasets required to optimize Power BI performance.
- Step 4: Integrate data sources into CSV databases importing them into PostgreSQL, defining the relational schema, and iteratively developing and validating SQL logics in DBeaver to produce the target datasets consumed by Power BI. All the curated sets were registered in “**datasets_generados**”
- Step 5: Automate dataset generation and updates using Python scripts, SQL statements, and executable processes, including:
 - Exporting Excel sources to standardized CSV files.
 - Loading CSV files into PostgreSQL and maintaining calculated columns through functions and triggers.
 - Executing SQL queries to generate curated fact and dimension datasets.
 - Connecting PostgreSQL-generated datasets to Power BI.
- Step 6: Adapt existing Power BI visuals and DAX measures to consume the optimized datasets, improving model performance, scalability, and maintainability.

With this revised approach, the project transitioned from a DAX-centric optimization strategy to a data engineering-oriented BI workflow, establishing a stronger foundation for long-term analytical scalability and performance.

Process – Analyze DAX and calculated columns & creation of “datasets_generados”

- The spreadsheet document “**datasets_generados**” was opened from Phase I folder, so could be adapted with new columns that improve the information gave.
- The column “*nombre_reporte*” was created informing to which report page corresponded each DAX measure used in Phase I. Unlike Power BI, where tooltip visuals were not separated into independent folders, in the Excel document they were documented separately. This allows measures to be duplicated without issue and assigned to multiple report pages when necessary.
- The visual Filtro por Fecha was used across most report pages in Phase I and is therefore classified under the “General” category, instead of being repeated on each individual page.
- The column “*usado_denuevo*” was updated to indicate how many measures were reused across visuals. At the end of Phase I, there were nine duplicated measures. Each measure was assigned a number per pair.
- As these changes affect the Phase I final documentation, the files were updated to reflect the amendments, ensuring they now contain the most up-to-date information.
- A new copy of the document was created to be considered for Phase II, named “**dax_mediciones_y_visuales_2**”.
- Using Power Query in Excel, new helping columns were added to the table to count the occurrences of the keywords “*analizar_CALCULATE*”, “*analizar_FILTER*”, “*analizar_IF*”, “*analizar_SWITCH*”, “*analizar_SUM*”, “*analizar_DIVIDE*” and “*analizar_ALL*” in each DAX measure. The original “*dax_formula*” column was replaced the values for a trim version.
- The column “*total_complejidad*” was created to sum all the occurrences found in the previous step. From here, the column “*estatus*” was created to inform all the measures that need to be analyzed. The formulas were grouped by:
 - *mantener*: The DAX measure is using less than two formulas to be considered, so they will remain with the same structure.
 - *revisar*: The DAX measure is using more than two and less than nine, so it will be reviewed to determine if an adaptation is needed.
 - *optimizar*: The DAX measure is using so many formulas that can be optimized, so these measures need to be restructured to be more efficient in Power BI.

- The column “*comentarios*” was created to inform if the measure will be affected for any table, or what will happen with it. If informs N/A, it means that the formula were considered for no changes
- To have a better distribution of the tables to be considered in this tables and for the generation of the curated datasets, the category table “*tablas_generadas*” was created. The document has the next columns:
 - *ds_id*: Unique id created to maintain an order in the table with the prefix CSV (it was assigned that name because the previous plan was to download the documents from Dbeaver as CSV format).
 - *sql_document*: informs the sql document where the SQL statement is saved
 - *tipo*: Informs which type of document referenced each table:
 - fact: Excel tables considered inside the main document of the Master Excel Documents. These documents are considered as the core tables of the project, as they have the data that feed the visualizations, measures and metrics in Power BI.
 - dimensional: Tables were created to complement the Fact tables in between the Master Excel Documents and are considered as reference/dimensional tables. Here are also included the tables created in PostgreSQL that works as dimensional tables, named with “dim_”.
 - dataset: Curated tables that improve DAX performance and support the visuals by complementing the fact tables.
 - helping: Tables created in PostgreSQL considered to help and feed a final dataset. These tables will not be considered in Power BI.
 - *reporte*: Informs to which report affect in the Power BI solution.
 - *nombre_tabla*: Name assigned to each document.
 - *tabla_origen*: informs from which document originated from each table. In the case of datasets where more than one table was needed to add column(s), the other table needs to be mentioned here.
 - *definicion*: Explains what is expected to obtain from the dataset.
 - *columnas_agregadas_sql*: Informs if a new column will be aggregated to the curated datasets after the code used in SQL.
 - *columnas_usadas_para_visualizacion*: Informs the columns originally used to create the datasets. The main goal is to have a more user-friendly version of the dataset used in Power BI. This tables can be view adding the prefix v_ before the name informed in column “*nombre_data*” (for example v_viaje_id) in Dbeaver.

- *columnas_usadas_para_powerbi*: Informs the columns that were used as resources of the Power BI solution, following the Star Schema design.

Process – Creation of “master_album_financiero”

As part of Phase II, the main document was renamed to “master_album_financiero” underwent the necessary structural adjustments and standardizations to support a scalable Business Intelligence workflow. These amendments were implemented to preserve the original financial logic while ensuring the data was consistent, normalized, and technically ready for downstream processing. Once these changes were completed and validated, the document was considered prepared to proceed with the creation of curated datasets in PostgreSQL, serving as the foundation for the Power BI solution and subsequent analytical layers.

1. Initial Setup & Table Structure Design

- From request of the stakeholder, a macro was created to filter out all the columns with the command Ctrl + Mayus + f.
- A macro was created to convert formulas in the “id” column in “c_diaria” to values with the command Ctrl + Mayus + i.
- For the “id” column in “c_diaria” spreadsheet, the letter A was added to the code (A000001).
- A new copy of the main document of the Master Excel Documents was renamed into “master_album_financiero”.
- In “c_diaria”, for all the transactions that were separated in two or more rows, they maintained their original code, but added a -01 (A000001-1, A000001-2, etc).
- In “c_diaria”, in “razon_uso” was decided to change the logic of category “P.VIAJE” where all the travels made (with an “id_viaje”) was considered under this category, even if wasn’t budgeted. Now there can be travels with id and with other categories.
- In “c_diaria”, in “clave” column, the next labels were merged:
 - “alimento” and “bebida” merged into “consumibles”.
 - “regalo” suffered a separation. All the transactions that didn’t correspond directly to a gift or donation were assigned into the new label “invitacion”.
 - “inversion” suffered a separation. All the transactions that correspond to a purchase or sale of a product, were assigned into the new label “comercio”.

- In “**c_diaria**” the columns “*m.extranjera*” and “*moneda*” were considered as helping columns. This were replaced by calculated columns obtained with SQL statements in PostgreSQL.
- In “**creditos**” the column “*fecha_inicio*” was created.
- In “**parametros**” the column “*Fecha Captura*” changed name into “*fecha_captura*”

2. “*id_presupuesto*” column

As part of Phase II, a standardized budget identifier “*id_presupuesto*” column framework was designed and implemented to enable consistent, scalable budget-to-actual analysis across the full historical dataset. The solution shifted budget tracking from a transaction-date dependency to a purpose-based analytical model, allowing spending to be evaluated accurately at the budget level regardless of payment timing.

By separating budget logic into a dedicated reference table and automating identifier assignment through the data pipeline, the model supports reliable aggregation, drill-through analysis, and long-term comparability in Power BI. This approach balances analytical accuracy, usability, and automation, ensuring that budget insights remain meaningful at both summary and transactional levels while keeping the semantic layer efficient and maintainable.

The procedures and design decisions applied during this phase are detailed in the following sections.

Expectations vs. Reality

The original recommendation was to create an “*id_presupuesto*” column in the Excel data source to:

- Track exactly how much was spent per budgeted item.
- Ensure month-to-month budget-to-actual alignment, so that delayed reimbursements or late payments do not distort monthly totals.

The approach was analyzed and adapted to the goals of this project:

- Create an “*id_presupuesto*” column for all specific budget items (“*razon_uso*” all the rows contain P.), informing the corresponding period of time (even if is longer than one month) and the total amount budgeted.

- Automatization of the “*id_presupuesto*” column with a new excel document that will work as a reference table informing the characteristics that need to follow each id, so they can be assigned automatically via PostgreSQL.

With this new approach, analysis by budget item will be easier for the user, as can see true spending per budget, regardless of transaction date, avoiding misinterpretation caused by late payments.

The chosen approach balances automation, clarity, and usability:

- Users can track spending per budget item accurately.
- Page-level visuals provide overview by month are possible in Power BI.
- Drill-throughs in Power BI reveal complete transaction details, even outside the filtered period.
- Some deviations from the ideal month-to-budget alignment are accepted as a reasonable trade-off to maintain simplicity and prevent over-complicated logic in Excel.

Creation of “*id_presupuesto*” and “*tabla_id_presupuesto*”

- In “**presupuesto**” spreadsheet, all the descriptions to be considered as different budgets, were separated with “,”, then they were separated into different rows & the ones corresponding to a same concept, were combined, so they can have a unique corresponding id for the whole corresponding period.
- A helping column “*id_presupuesto*” was created in “**c_diaria**” in order to audit the transactions assigned to each specific budget (informed with “P.” in “*razon_uso*” column. This column will be deleted as soon as the audit is done with the curated dataset generated in PostgreSQL.
- The column “*id_presupuesto*” was created in the spreadsheet “**presupuesto**”.
- The reference table “**tabla_id_presupuesto**” with the purpose of having in one table, all the ids used over the past 10 years and serves as reference table to “**master_album_financiero**” document.
- The first section of the table “**tabla_id_presupuesto**” focused to understand to what corresponds each id created.
 - “*Id_presupuesto*”: informs the corresponding id that will be considered per transaction in “**c_diaria**” spreadsheet.
 - “*concepto_principal*”: informs the main concept that will refer the id in reference.
 - “*tipo*”: There will be two types. “recurrente” corresponds to the ids that will be appearing over the years. “unico” is referenced only for a

specific event, mainly this will be established for travels that have “P.VIAJE” as “razon_uso”.

- “*descripcion*”: Informs a description of the referenced id.
- The second section of the table “*tabla_id_presupuesto*” focused to give the specifications that need to have the transaction so can be assigned to an id automatically. If the id does not have a value in any column, that means can have any value in it:
 - “*grupo_viaje*”: required field for all the “tipo” unico. Assigns the corresponding travel group for each id created.
 - “*razon_uso*”: Required field. The corresponding concept used for each id. When the id could be more than one, will be informed as “P.” If this field is in orange, it means is the last logic in the if statement (if all the rules didn’t apply, then the corresponding id will be utilized).
 - “*clave*”: The corresponding label used per id. If two or more labels can be used, then this field remains without information.
 - “*descripcion_c_diaria*”: Informs if the id has a specific text to help identify it in the data source. If there is no specific text, remains in blank this field.
- For general situations, the following rules were established to assign a corresponding budgeted id:
 - The guiding principle is purpose alignment. Codes are not assigned mechanically but based on whether the budget represents a recurring theme or a unique event.
 - This ensures that comparisons in Power BI remain meaningful, avoiding both unnecessary fragmentation (splitting one theme into too many codes) and excessive lumping (merging distinct events into one code).
- For Recurring/Thematic Budgets that appear throughout the years in the document (e.g., hobbies, projects, subscriptions), the next rules were established to assign a corresponding budgeted id:
 - This will be referenced as “recurrente” in column “*tipo*”.
 - A single budget ID will be assigned and reused across multiple months and the years. For example, adequations to the house, all purchases related to this theme, even if they occur in non-consecutive months or different years, will share the same code.
 - The purpose is to measure how much has been budgeted and spent in total on the same theme over time.
- For Event-Based Budgets (e.g., trips, one-off projects), the next rules were established to assign a corresponding budgeted id:
 - This will be referenced as “unico” in column “*tipo*”.

- Each event will have its own unique budget ID. For example, A travel made to a same place in February 2024 and October 2024 for different reasons, will each have a distinct code, even if the location is the same.
- The purpose is to isolate spending per trip/event, while still allowing analysis by higher-level categories (city, country, continent) in Power BI.
- When a transaction can be assigned in two “*id_presupuesto*” and they are in different “*tipo*” references, the next options will be considered:
 - If the transaction falls between the first eight “*id_presupuesto*” (P001-P008), the id will be considered between that range.
 - If the transaction is outside of the range previously informed, it will be considered as corresponds.

3. “*id_deuda*” column

As part of Phase II, a structured debt identification model was implemented through the creation of the “*id_deuda*” column and its corresponding reference table. This approach enabled consistent tracking of pending and settled “deuda” transactions across the dataset, supporting clearer visibility of outstanding obligations in the Power BI solution. By standardizing how debts are identified, described, and referenced, the model improves analytical clarity, allows accurate aggregation and filtering in Power BI, and ensures that pending payment insights remain reliable over time without increasing complexity in the report logic.

The procedures and design decisions applied during this phase are detailed in the following sections.

Expectations vs. Reality

The original recommendation was to create an “*id_pendientes*” column in the Excel data source to:

- Track pending payments for “**deudas**” transactions more precisely.
- Ensure the Dashboard **Resumen Financiero** can show which purchases are still pending or already settled.

The approach was analyzed and adapted to the goals of this project:

- An “*id_deuda*” in “**c_diaria**” was created. The ID will be generated when “*clave*” = “deuda” (e.g., D001, D002...).
- If the debt was paid by other mean that aren’t monetary, the special id “D000” will be assigned.

Creation of “*id_deuda*” column

- In “*descripcion*” column, the third-party name was adapted to always appear after the first “:”.
- For the descriptions that started with “A meses:”, the phrase was moved to the end of the text for consistency.
- For the transactions made between an “*id_deuda*” that doesn’t correspond to add a name after “:”, they will be considered as part of the debt to the person who gave more money for the whole transaction (excluding the user).
- The reference table “*tabla_id_deuda*” was created as a reference table, so each “*id_deuda*” created could be found with their respective description for the easiest understanding of each id. The document has the next columns:
 - “*id_deuda*”: The corresponding id created in “*c_diaria*”
 - “*descripcion*”: Informs a short explanation of the meaning of each debt registered.
 - “*fecha_inicio*”: Informs the date were the corresponding id appeared in “*c_diaria*”
 - “*finalizado*”: The user declares if the debt can be considered as closed with “si”, or if not, will remain empty. For the id D000 will be assigned N/A.

4. Standardization of name extraction in column “*descripcion*”

The decision to proceed with the identification and standardization of all third-party names in the “*descripcion*” column from the spreadsheet “*c_diaria*” was made after the successful implementation of the “*id_presupuesto*” column. This new standardization step aims to create a universal third-party dimension table, allowing the Power BI solution to generate a dynamic, centralized report focused on all individuals referenced throughout the “*master_album_financiero*” document.

By cleaning and extracting all valid names, this process ensures consistent, analyzable data across categories such as debt, gifts, invitations, reimbursements, and any future classifications. It also eliminates ambiguous or inconsistent text patterns, making the third-party field reliable for slicing, filtering, and drill-through analysis.

Implementing this process not only strengthens the data model but also enables new analytical perspectives, such as understanding spending patterns by person, aggregating historical interactions with third parties, and providing a unified view that supports both detailed and high-level reporting within the Power BI ecosystem.

The next rules were considered in the standardization of the column:

- All the third parties that are considered as names, will be appearing at the end of the text after the colon ":". (For Example, Gasto: Jose Lopez).
- Only the cases applying to the first rule will appear after a colon ":". For this matter, the next cases suffered changes:

- The special cases required in table **"id_presupuesto"**, in the column **"descripcion_c_diaria"**, which the colon was replaced with a dot "."

Example: "Experiencia: Evento" → "Experiencia. Evento".

- For cases where the transaction corresponds to deferred months for credit payments, they were replaced with a dot "." before the text "A meses".

Example: "Gasto. A Meses: Jose Lopez". → " Gasto: Jose Lopez. A Meses"

- Any colon that does not indicate a third-party's name (based on the first rule) is removed.

Example: "Auto: Nissan" → "Auto Nissan".

If the colon is part of an original title or proper name, it is replaced with a hyphen ("-").

Example: "The Avengers: End Game" → "The Avengers – End Game".

5. Creation of folder for images of travels

During Phase II, travel-related enhancements were evaluated with a focus on improving analytical consistency and user experience in Power BI. While full automation of the "desde" and "hasta" columns was not feasible due to historical transaction timing constraints, the limitation was documented to preserve data integrity. In parallel, a visual enrichment strategy was successfully implemented by linking trip identifiers to external image resources through a dedicated reference table. This design enables richer contextual insights in the **"Reporte Viajes"** page without impacting data model performance, demonstrating a balanced approach between analytical rigor and user-oriented storytelling.

The procedures and design decisions applied during this phase are detailed in the following sections.

Expectations vs. Reality

- Automate **"desde"** and **"hasta"** columns in the **"viaje"** table, to reduce manual errors in Excel and standardize data input in Power BI.

- Enhance tooltips in the **“Reporte Viajes”** page by including a photo took from each travel (when available), enriching the personal experience.

The automation of the *“desde”* and *“hasta”* columns was not feasible at this stage because the **“master_album_financiero”** records include all purchases related to each trip, many of which occurred before the official start date of the trip. For the second recommendation, the folder **“powerbi_viajes”** was created and the corresponding images were saved there to proceed, and those documents will be related in the new table **“viajes_url”**.

Creation of folder **“powerbi_viajes”** & creation of table **“viajes_url”**

- The folder **“powerbi_viajes”** was created as expected. All trip photos are saved in one folder, and each image was renamed after the corresponding *“id_viaje”* (V001.jpg, V002.jpg). For trips without an available photo, a generic image of the destination was used.
- After evaluating several methods to automate the creation of the **“viajes_url”** CSV file, the chosen solution was to upload the images to GitHub. The resulting CSV document contains the following columns:
 - *“image_url”*: the public URL generated for each image.
 - *“id_viaje”*: the corresponding trip ID assigned to each image.

6. Review of Historical Data Availability for **“creditos”** spreadsheet

Expectations vs. Reality

- Analyzed it is possible to add all the past periods of time to the **“Reporte Creditos”** tab, this to have available the most complete information about the credit debts.

The approach was analyzed and adapted to the goals of this project:

- With the help of previous versions of the **“master_album_financiero”** spreadsheets, the corresponding *“fecha_corte”* and *“fecha_limite_pago”* for earlier periods were reconstructed without difficulty. This was possible because each transaction was consistently recorded under the concept *“pago: tarjeta de credito”* with the exact amounts that had been settled.
- For the transactions where other credit *“metodo_pago”* were used, it was decided to define the same structure of months, starting from the seven, eight or nine days of the month and ending between twenty-nine to thirty-one, leaving the *“fecha_limite_pago”* between the last two days of the corresponding month, or the first day of the consecutive month.

Process – Data sources integration and datasets creation

We proceeded with the data loading into PostgreSQL to create the database we will require to create the curated datasets. We took the data source “master_album_financiero” and the reference/dimensional tables and converted them into tables in PostgreSQL, where each spreadsheet appeared as one table.

- The primary key is defined with the corresponding column with unique values per row (id columns and “ubicacion_viajes”) assigned per table in Excel. If a table didn’t have a unique id, a numeric id was assigned with a SQL statement.
- A new database “album_financiero_bd” was created in PostgreSQL, where all the tables loaded are maintained under one structure. This helps to have them organized and grouped in one place, instead of remaining in public.
- The database “album_financiero_bd” was uploaded to DBeaver, where a diagram was created to validate that each table was created correctly:
 - The Primary Keys (pk) have been assigned correctly.
 - The columns have the correct type assigned.
 - The consistency on the names.

Process - Creation of datasets in PostgreSQL via DBeaver

- The folder “SQL Scripts” was created, where they are separated in three categories divided by numbers:
 - Scripts starting with 1. correspond to the creation of the PostgreSQL database, the import of the master source documents, and the generation of the base tables used by the Power BI solution. This group also includes the scripts used to automate table population and data updates.
 - Scripts starting with 2. correspond to the generation of curated datasets and dimension tables required for the Power BI data model to function as expected.
- The document “tablas_generadas” was saved in the same folder. This document was used to register and maintain the control of all the imported databases into Dbeaver, the organization of each curated dataset generated in PostgreSQL and the columns each dataset have been assigned with SQL statements.
- Two versions of each curated dataset generated in Dbeaver were created:

- Curated datasets version to build enriched tables using JOINS between fact tables and dimension tables. These versions shows all the columns and logic required to validate calculations but does not store data physically. The main goal is to have a more user-friendly version of the datasets used in Power BI in case the user needs a better understanding of these tables and can view them in Dbeaver. These tables are saved with the prefix `v_` (for example `v_ds_credits_periodo`).
- The final curated datasets extracting only the necessary fields required for Power BI to create the corresponding relationships between tables in Model View. The Primary Keys are assigned in this version.
- All the columns created as periods of time with the code YYYYMM, they were considered as INTEGER in the SQL statements.
- The next rules were created to give a hierarchy to be followed for the SQL statements, so the automatically assignation of the *“id_presupuesto”* to the corresponding dataset was implemented as expected. When a rule is completed, the transaction is not considered for the remaining rules. For example, if the transaction matches with Rule 1, the SQL statement will finish with the row and continue with the next transaction.
 - Rule 1. Match by *“descripcion_c_diaria”*: If the column *“descripcion”* partially matches with the concept informed in *“descripcion_c_diaria”* in the reference table **“tabla_id_presupuesto”**, the reference id will be assigned. This is highest priority. If there is no match, we proceed with the next rule.
 - Rule 2. Match by *“grupo_viaje”*: compare between the columns *“grupo_viaje”* of each table. If there is an exact match with an existent concept defined in **“tabla_id_presupuesto”**, the reference id will be assigned. If there is no match, we proceed with the next rule.
 - Rule 3.1. Match by *“razon_uso”*: compare between the columns *“razon_uso”* of each table. After finding an exact match, the next rule will follow.
 - Rule 3.2. Disambiguate by *“clave”*: compare between the *“clave”* columns of each table. Then proceed in order looking to the next statements:
 - If there is an exact match, the id is assigned.
 - If there is no match and the column *“clave”* in **“tabla_id_presupuesto”** is empty, assign the remaining transactions with that *“razon_uso”* that haven't been assigned yet to an id.

- The helping column “*id_presupuesto*” created in the spreadsheet “**c.diara**” in the “**master_album_financiero**” table was deleted after the confirmation of this SQL statement works as expected.

Automatization process – Python Script “actualizar_album_financiero.exe”

One of the main goals requested in Phase II was the automation of uploading all information registered in the Master Excel documents into the Power BI solution. To achieve this, several scripts and processes were developed, resulting in a well-organized structure combining Python automation, SQL statements, and PostgreSQL enhancements through DBeaver. All these elements were organized into a set of dedicated EXE tools—each responsible for a specific part of the workflow—to keep the entire system easy to test, maintain, and understand. Together, these EXE documents allow the user to update all Excel data sources and the “**powerbi_viajes**” image folder with just one click.

Python was used as an orchestration layer to automate data reloads and ensure synchronized updates between the Excel data sources, PostgreSQL (via DBeaver), and downstream Power BI datasets. This approach minimizes manual intervention and guarantees consistent analytical snapshots across the project.

All the EXE documents were unified in the master document “**actualizar_album_financiero.exe**”. The scripts used to automate each section explained below and the final script used to group them in one EXE document, were saved in the folder “**Python Scripts**”.

While the entire backend automation is now fully streamlined, the automatic refresh and publication of the updated data in Power BI require a paid Power BI license. Without one, these final steps must still be performed manually inside Power BI Desktop.

Python updating Script “actualizar_powerbi_viajes”

To automate the management of the travel images used in the folder “**powerbi_viajes**”, a Python script was created that detects modifications in the local image directory, updates the corresponding reference file “**viajes_url**”, and synchronizes all changes with GitHub without requiring user intervention.

To ensure accurate change detection, the script generates and stores a unique hash code for each image, allowing it to identify modifications even when filenames remain the same. Throughout the process, it displays clear progress messages, so the user always knows which step is being executed.

Python Exporter Script “exportar_a_csv”

To streamline the generation of clean CSV files from the Master Excel Documents, a Python-based automation workflow was developed. The script followed the next considerations before the creation of each CSV datasource:

- The syntax of the date columns must be separated only by slashes (01/01/2021) and not include any text.
- Copy as static values all the columns with formulas.
- Cleaning currency columns by removing symbols (\$) and thousand separators.
- Standardization of all the text column, erasing all the double quotes (”) and commas (,). This to avoid issues of readiness when the CSV document is uploaded into PostgreSQL program.
- Save the document as CSV UTF-8 (Comma delimited) (*.csv) as the document have special characters like ñ and accents.

Python CSV to PostgreSQL Script “actualizar_tabla”

To streamline the data-loading process for the Master Excel Documents, the Python automation “actualizar_tabla” script was developed to import the csv datasources generated with “exportar_a_csv”. The script removes the need for manual data imports in DBeaver and ensures that all fact and dimension tables remain synchronized with the most recent CSV datasources.

As part of the implementation, additional tables such as “tabla_tipo_cambio” and “viajes_url” were integrated into the process. A mismatch in dictionary keys was identified and resolved, ensuring full alignment between the script’s configuration and the database schema.

Creation of functions and triggers in PostgreSQL

To ensure the databases tables remain fully synchronized with the updated Excel source files imported using “actualizar_tabla” and calculate the corresponding columns created in Dbeaver, an automated calculation layer was implemented directly in PostgreSQL using custom functions and row-level triggers with AFTER INSERT OR UPDATE SQL triggers and functions.

This approach removes the need to manually run UPDATE statements after each data load and guarantees consistency, accuracy, and automation across all fact_ and dim_ tables involved in the star-schema model (the tables coming from the

Master Excel Documents). It also aligns the project with professional data-warehouse standards by embedding business logic directly into the database layer.

Curated datasets in PostgreSQL and Python script “refrescar_tablas”

To be able to automatize the complete process done in PostgreSQL and after analysis, the decision to change the original approach:

- from using CREATE VIEW for flat tables (informed as v_ds_ tables) and CREATE TABLE for schema tables (informed as ds_ tables and used in Power BI) with CREATE MATERIALIZED VIEW, this to improve performance, simplify maintenance and help with the automatization of the creation of the curated datasets.

Materialized views tables solve both issues by storing precomputed results and allowing controlled refreshments. This creates a faster, more predictable, and easier-to-maintain transformation layer between the databases in PostgreSQL and Power BI. The new structure ensures that all curated datasets are consistent, optimized, and automatically updated during the daily pipeline.

To ensure all curated datasets remain synchronized with the latest data, the “refrescar_tablas” script was created that automatically refreshes every materialized view in the schema “album_financiero_bd”. Instead of relying on scheduled jobs or manual execution, the script runs as part of the daily ETL process—immediately after the Excel datasets are imported into PostgreSQL and the triggers have updated the information into the databases.

The script connects to the database, detects all materialized views dynamically, and refreshes each one in sequence while logging clear progress messages. This approach guarantees that every dataset feeding Power BI is always up to date, removes the risk of forgetting to refresh a specific view, and simplifies the entire update pipeline.

Process – Connection between PostgreSQL (via Dbeaver) and Power BI

At the initial stage of Phase II, curated datasets were imported into Power BI using CSV files exported from PostgreSQL. This approach was intentionally selected to validate schema design, column formats, data standardization rules, and overall analytical logic before introducing full database connectivity. Using static files at this stage reduced complexity and ensured that the data model behaved as expected within Power BI.

Once the datasets were validated and considered structurally stable, the solution evolved to a direct connection between PostgreSQL and Power BI. This transition reduced manual intervention, improved refresh performance, and aligned the solution with a production-oriented BI architecture, where Power BI consumes data directly from a centralized database layer.

During this transition, a platform-specific limitation was identified: Power BI does not automatically list PostgreSQL materialized views in its Navigator interface. To integrate these views into the semantic model, they were accessed by explicitly defining SELECT queries within the Power BI connection. This approach ensured full compatibility with the curated datasets while maintaining the intended database-driven design.

Process - Upload and defining relationships between tables in Power BI

The relationships between tables were created in Power BI Model View instead of PostgreSQL (DBEaver). This decision allows greater flexibility for structural adjustments, such as adding, removing, or modifying connections between tables, without the need to alter the database schema in DBEaver. This approach ensures faster iteration, reduces the risk of conflicts during data loading, and keeps the PostgreSQL database optimized as a clean data source rather than a rigid relational system.

As the structure of the new SQL-based datasets differs from the original Excel sources, a new Power BI report was created from zero called “[master_album_financiero](#)”. This approach ensures full consistency between the PostgreSQL database and the Power BI data model, avoids compatibility issues caused by renamed or normalized columns, and reflects the structural improvements achieved in Phase II of the project.

Additionally, the new model design significantly reduces Power BI’s workload by performing most transformations, cleaning steps, and normalizations within DBEaver and PostgreSQL. Instead of flattening datasets, the model now follows a relational approach, maintaining key-based connections between fact and dimension tables. This strategy minimizes data redundancy, ensures higher consistency across all datasets, and results in a more scalable, efficient, and easier-to-maintain Power BI architecture.

The next rules were created for the relationship of the tables in Power BI:

- In Power BI > File > Options > Time Intelligence, the function Auto Date/Time for new files was disabled.
- In Power BI > File > Options > Relationships, all the functions were disabled.

Fact Tables:

- Fact tables represent transactional, historical or metric data and are informed in the document “**datasets_generados**” in the column “*tipo*” as fact. Their name has the prefix fact_ before the name of the document.
- Every fact table that contains a date must connect directly to the “**Calendario**” table created in Power BI with a DAX measure.
- Relationship direction: “**Calendario**” -> Fact table (Single filter direction).
- Never directly relate “Fact table” <-> “Fact table”. Facts only relate through shared Dimensions.

Dimension Tables:

- Dimension tables provide descriptive attributes used by Fact tables (example: payment method, investment type, geography, type of budget, type of debt, type of trip). They are informed in the document “**datasets_generados**” in the column “*tipo*” as dimension. They have the prefix dim_ before the name of the document.
- Relationship direction: Dimension -> Fact (Single filter direction).
- Dimension columns are not expected to repeat inside the dimension itself, but they can repeat inside Facts.

Datasets created Dbeaver with SQL :

- Curated documents obtained that will serve as datasets to complement information on the Fact Tables. They are informed in the document “**datasets_generados**” in the column “*tipo*” as dataset. They have the prefix ds_ before the name of the document.
- Relationship direction: Can connect to Fact tables with single direction relationship cardinality must respect actual grain (one-to-one or many-to-one). Dataset -> Fact table (Single filter direction).
- Relationship direction: Can connect to Dimension tables using many-to-one single direction. Dimension tables -> Dataset.
- Datasets cannot create direct relationships between Fact table <-> Fact table. They can relate to multiple Fact tables only if they serve as an intentional bridge table with one-direction relationships that respect grain and do not bypass dimensions.

- Only connect to “**Calendario**” table if they contain independent date logic (not inherited from the Fact table source).
- These datasets complement the model but do not replace the main Fact table.

The document “**relaciones_tablas_powerbi**” was created with the program DAX Studio, to have a well-organized spreadsheet to easily find the relationships made in the Model View in Power BI.

Analysis

In Phase I, budgeting and allocation logic was implemented entirely within a single DAX measure. This approach required dynamic row-by-row evaluation, date comparisons, string parsing, and iterative calculations each time a visual was refreshed. While functionally correct, this design increased computational load and limited scalability as data volume and report complexity grew.

In Phase II, this logic was migrated upstream into curated datasets generated in DBeaver. By materializing and organizing the data structure at the database level, Power BI was able to consume pre-processed, analysis-ready tables. This significantly reduced the computational burden on the semantic layer, resulting in faster refresh times, improved performance, simplified DAX measures, and greater maintainability for future expansion.

Certain transformations, such as the concatenation of multiple columns to create descriptive labels for visuals, were intentionally retained within Power BI measures. These calculations are presentation-focused and do not impact data aggregation logic, making them more suitable for handling at the semantic layer.

As this project represents a continuation of Phase I, the analytical structure of the Power BI solution follows the same conceptual framework. Page layout logic, visual organization, template usage, and the definition of key questions per report page were preserved and adapted to align with the updated data model and objectives introduced in Phase II.

Additionally, several period-based integer columns (YYYYMM), prefixed with “period_”, were created within the SQL curated tables to practice data warehouse modeling techniques and simulate optimizations commonly used in large-scale environments. During the Power BI modeling phase, these columns were evaluated alongside date fields. While period columns are valuable for database performance and partitioning strategies, the Power BI semantic model relies on date columns to support time intelligence functions, slicer hierarchies, and relationship behavior. As a result, date fields were retained for analytical calculations and visualization, while period columns remain available for reference and future scalability.

Distribution between the Power Bi Solution

The Power BI solution was structured around two dashboard-style pages and multiple report pages. One dashboard focuses on the user’s current financial status, while the second acts as a summary hub for historical analysis (both are explained in detail in the Share phase). The report pages are designed to answer the key questions redefined for Phase II of the project.

The following main tables and visuals were created as part of the Power BI solution:

- A calendar table named **“Calendario”** was created to function as filter for slicers.
- Filtro por Fecha: Slicer created in all the pages, to filter the visuals by date, as required by each page (year, quarter, month and/or day).
- Search bars created to be able to filter by text columns in the tables “Detalle de transacciones”. A bar is created per column that the user will require to look up a specific concept.

In the following section, each report page created in the solution is presented. For every page, the sections, visuals, and metrics are explained, along with the DAX measures used to support each visual and calculation.

Where has my money gone over the years? Which concepts and labels represent the highest spendings?

A report page called **“Reporte Cuentas Diarias”** was developed to provide a comprehensive view of all transactions recorded over the past 10 years within the **“fact_c_diaria”** table. This page enables users to explore complete historical records and analyze long-term spending patterns across concepts, labels, and usage types, supporting both detailed transaction review and high-level behavioral analysis.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Totales de cuentas diaras:** Section created to inform the total amounts of the master document by the selected period.
- **Card visual:** Inform the gross net obtained from all the transactions, the real net and the transactions that weren't possible to locate to which operation corresponded, by the selected period.
 - Total Cuentas Diarias: Calculate the total amount of the master document by the selected period.
 - Total Neto: Calculates the real amounts that generated gains or losses in the master document, excluding the transactions where **“razon_uso”** = DINERO and **“clave”** = “transaccion”.
 - Total DINERO y Transaccion: Calculates the amount generated by the transactions where **“razon_uso”** = DINERO and **“clave”** = “transaccion”.
- Line chart: Displays the total amounts of the master document over the selected period.
 - Total Cuentas Diarias.
- **Top 10 gastado por razon de uso:** Section created to inform the top 10 higher expenses made by concept **“razon_uso”**

- Slicer visual: A dropdown visual created to select a corresponding concept existent between the selected period.
 - Cuentas Diarias Visible: Inner filter created to inform only of the transactions that are considered between the selections made between the page.
- Clustered bar chart: Displays the total amount of the transaction made by the selected concept.
 - Total Cuentas Diarias.
- **Magnitud de gastos por clave:** Section created to show the magnitude of transactions made on the master document, by the selected period.
- Slicer visual: A dropdown visual created to select a label, as informed in “clave” in the table “fact_c_diaria”.
- Treemap: Displays the magnitude of each transaction made by label between the selected period of time, which higher the total made per label, bigger the section assigned will appear in the visual.
 - % Magnitud Cuentas Diarias: Informs in percentages the magnitude of each label appearing in the selected period.
 - Magnitud Cuentas Diarias: calculates the magnitude of each label appearing in the selected period. ABS() is used to represent the magnitude of financial movements in the visual, ensuring that expenses stored as negative values are visualized proportionally by size without sign-related distortion.
 - Total Cuentas Diarias.
- Detalle de transacciones: Displays detailed transactions filtered by selections made in the summary visual.
 - Total Cuentas Diarias.

What value has been generated by third-party reward programs? After all the money I deposited and withdrew, how much did the investment actually generate? Since the start of the investment report, what amount of investment capital is currently available, and how is it being used over time?

The report page “**Movimientos Inversiones Disponibles**” was developed to analyze how available investment balances have evolved over time. This page focuses on transaction-level movements that directly impact the total amount available for use, including deposits, withdrawals, and balance-affecting adjustments.

Third-party reward programs are incorporated into this report following the same analytical treatment applied in Phase I. Costs associated with the use of financial

products (such as annual fees or related charges) are classified as deposits that reduce available balances, while rewards received are treated as withdrawals or returns, reflecting increases in available investment funds over time.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Tendencia en inversiones disponibles y monedero:** Section created to display the investments totals by the selected period.
- **Slicer Visual:** Created to select all the deposit or withdrawals (“deposito” or “retiro”) made over the selected period, as informed in the column “*categoria_movimiento*” in the table “*dim_categoria_movimiento*”
- **Line chart:** Shows the investments amount tendency over the selected period.
 - **Total Inversiones Monedero:** Calculate the generated value of investments over the selected period.
- **Card visual:** Shows the investments amount tendency over the selected period.
 - **Total Inversiones Monedero.**
- **Desglose de inversiones disponibles y monedero:** Section created to inform how the investment selected changed over the selected period.
- **Waterfall chart:** Show the deposits and withdrawals of the selected available investment, over the selected period.
 - **Total Inversiones Monedero.**
- **Card visual:** Informs the total amount of the deposits and withdrawals of the selected investment, by the selected period.
 - **Depositos Inversiones y Monedero:** Informs the total amount of the deposits of the selected investment, by the selected period.
 - **Retiros Inversiones y Monedero:** Informs the total amount of the withdrawals of the selected investment, by the selected period.
- **Tipo de inversiones:** A dropdown created to select the required investment. Each investment informed in the column “*tipo_inversion_detectado*” in the table “*dim_inversiones_movimientos*”.
- **Detalle de transacciones:** Displays detailed investment-related transactions filtered by selections made in the summary visual.
 - **Total Inversiones Monedero.**
- **Historico balance final de inversiones:** Informs the total amount generated by all the existing investments in the selected period. This visual works as Drill-Through to “**Analisis de Inversiones**”.
 - **Balance Final en inversiones:** Calculate the total amount generated in the selected period.

A report page “**Análisis de Inversiones**” was developed to analyze how investment balances have evolved over time, influenced by both internal movements (deposits and withdrawals) and external market fluctuations.

As documented in Phase I, the “**Inversiones**” analysis was built using the information available at the time, which resulted in incomplete movement detail for certain investments. This limitation is particularly evident for records classified under the category “no disponible,” where transaction-level data is not fully captured.

To address these gaps, several calculations were initially designed to estimate the impact of market behavior between reporting periods. However, because the investment dataset is structured as periodic monthly snapshots rather than granular transactional records, key measures—such as initial balances and accumulated impacts—proved unreliable when implemented solely through DAX logic.

As a result, the scope of this analysis was constrained to begin in 2022, the most recent period in which monthly snapshots could be consistently standardized, ensuring analytical reliability and interpretability within the Power BI model.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Saldo acumulado de inversiones a lo largo del tiempo:** Section created to display the values of the investment appearing in the selected period.
- Slicer visual: A dropdown visual created to select the required investment. Each investment informed in the column “*tipo_inversion*” in the table “*dim_tipo_inversion*”
 - Tipo Inversion Visible: Inner filter that shows only the available investment in the selected period.
- Table visual: Displays the current value of each investment in the selected period.
 - Balance Final en Inversiones.
 - Balance Inicial en Inversiones: Return the snapshot balance per investment type based on the last available month previous the selected period.
- Line chart: Shows the comparative between the opening balance & the final balance trending changes over the selected period.
 - Balance Inicial en Inversiones.
 - Balance Final en Inversiones.
- **Totales balances por tipo de acceso:** Section created to inform the final totals obtained from the balances, by the selected period.
- Slicer visual: A drop created to select the type of investment (available if the investment can have withdrawals between its existence, or not available if the

investment is accumulative and can be withdrawn in the future). This is informed in “*tipo_acceso*” in table “*dim_tipo_inversion*”

- Card visual: Display the variation between balances on the selected period.
 - Balance Inicial en Inversiones.
 - Balance Final en Inversiones.
 - Diferencia de Balances: Calculates the difference between Balance Final en Inversiones, minus Balance Inicial en Inversiones.
 - Balance Cambio % con Flecha: Show the difference between balances in percentages with a symbol showing if the value is negative or positive.
 - Diferencia de Balances en %: Calculates the difference between balances in percentages by the selected period.
- **Metricas:** Section created to show the final metrics obtained from the balances in the investments selected over the period.
- Stacked column chart Resultado neto de la inversion: Shows and informs the difference between Valor neto generado final and Valor neto generado inicial.
 - % Valor Neto Inicial vs Resultado Neto: Informs the Valor Neto Generado Inicial amount compared with the Resultado Neto de la Inversio in percentages.
 - Resultado Neto de la Inversion: calculates the total amount obtained from the difference between the most recent and first snapshot balance appearing in the selected period.
 - % Valor Neto Final vs Resultado Neto: Informs the Valor Neto Generado Final amount compared with the Resultado Neto de la Inversio in percentages.
 - Resultado Neto de la Inversion.
- Gauge visual Valor neto generado inicial: Shows and informs the first existing value informed by the selected period.
 - % Valor Neto Inicial vs Resultado Neto.
 - Valor Neto Generado Inicial: Return the snapshot balance per investment type based on the first available month of the selected period.
- Gauge visual Valor neto generado final: Shows and informs the most recent existing value informed by the selected period.
 - % Valor Neto Final vs Resultado Neto.
 - Valor Neto Generado Final: Return the snapshot balance per investment type based on the most recent available month of the selected period.

What is my total monthly budget? How much remained unspent? How is it allocated across each “*id_presupuesto*”?

A report page titled “**Reporte Presupuesto**” was developed to analyze budget-related transactions over time. This page focuses on detailed movements associated with each budget, enabling comparison between actual spending and the budgeted amounts defined by the “*razon_uso*” column in the “*fact_c_diaria*” table. It provides visibility into how funds have evolved over time, supporting both period-level monitoring and long-term budget performance analysis.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total gastado por presupuesto:** This section inform the comparative between the budgeted amount vs. the actual spending made over the selected period.
- Card visual: Informs of the budgeted amount, total spent and the remainder of the selected period.
 - **Monto Gastado:** Calculate the sum of the total transactions made by the selected period.
 - **Monto Presupuestado:** Calculate the sum of the total budgeted amount filtered by the selected period.
 - **Presupuesto Especifico:** Calculate the sum of the specific budgets informed in “*tipo_razon_uso*” in the table “*dim_razon_uso*”.
 - **Presupuesto Regular:** Calculate the sum of the regular budgets informed in “*tipo_razon_uso*” in the table “*dim_razon_uso*”.
 - **Restante para gastar:** Calculates the remaining amount pending to be used from the budgeted amount. If it appears green, it means that the user spends less money than budgeted. Red means that the user spent more money than budgeted.
- Area chart: Displays the historic totals of regular budgeted transactions over the years.
 - Monto Gastado.
 - Monto Presupuestado.
- Gauge chart: Displays in percentages how much budget was used compared to the budget issued over the selected period.
 - % Presupuesto: Informs in percentages how much was used compared to the budget issued over the selected period.
- **Comparativa presupuesto por razon de uso:** Section created to see the comparative between the budgeted amount vs. the total spending made by the selected period.

- Slicer visual: Created to select for type of id budget defined as “especifico” for all the budgets created in a defined month and “regular” that occurs for periods between two months and a complete regular year.
- Clustered column Chart: Displays the comparative between the budgeted amount vs. the total spending made by their corresponding concept used in their corresponding “razon_uso” column, with the help of the “dim_razon_uso” table, for the selected period.
 - Monto Gastado.
 - Monto Presupuestado.
- **Top 5 gastado por clave:** Informs the top 5 expenses made by label by the selected period.
- Slicer visual: A dropdown visual letting the user to select a value by level, as informed in their corresponding column “clave”, with the help of the table “dim_clave”
- Card visual: Informs the percentage corresponding of the top 5 labels with higher expenses in the selected period.
 - % Monto Gastado: Calculates the percentage corresponding of the selected label(s).
- Clustered bar chart: Informs the distribution between the top 5 labels with higher expenses in the selected period.
 - Monto Gastado.
 - % Monto Gastado.
- **Historico especifico ID presupuesto:** Section created with a card visual, displaying the sum of the total budget created specifically for a month, meaning they have an assigned id budget. This visual works as Drill-Through to “**Reporte ID Presupuesto**”
 - Monto Gastado por ID: Calculate the sum of the total specific budget amount.
- **Detalle de Transacciones:** Displays detailed budget-related transactions per row based on selections made on the report page.
 - Monto Gastado.

A report page called “**Reporte ID Presupuesto**” was developed. This page focuses on detailed transactions informing all the movements made as specific budgets per assigned budget id, as informed in “id_pendiente” column in the tables “fact_presupuesto” and “tabla_id_presupuesto”.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total gastado por presupuesto:** This section inform the comparative between the budgeted amount vs. the actual spending made over the selected period.
- Card visual: Informs of the budgeted amount, total spent and the remainder of the selected period.
 - Monto Gastado por ID.
 - Presupuesto Especifico.
 - Restante para gastar por ID: Calculates the remaining amount pending to be used from the specific budgeted amount. If it appears in green, it means that the user spends less money than budgeted. Red means that the user spent more money than budgeted.
- Area chart: Displays the historic totals of specific budgeted transactions over the years.
 - Monto Gastado por ID.
 - Presupuesto Especifico.
- Gauge chart: Displays in percentages how much budget was used compared to the budget issued over the selected period.
 - % Presupuesto Especifico: Informs in percentages how much was used compared to the specific budget issued over the selected period.
- **Comparativa presupuesto especifico por razon de uso:** Section created to see the comparative between the budgeted amount vs. the total spending made by the selected period.
- Clustered column Chart: Displays the comparative between the budgeted amount vs. the total spending made by their corresponding concept used in their corresponding “razon_uso” column, with the help of the “dim_razon_uso” table, for the selected period.
 - Monto Gastado por ID.
 - Monto Presupuestado.
- **Top 10 gastado por ID:** Section created to inform the top 10 expenses made by id budget by the selected period.
- Slicer visual: Created to select between “recurrente” that are current IDs assigned over the whole period or “unico” that are unique IDs assigned for no repetitive events occurring in a specific period, as informed in the column “tipo” in the table “dim_id_presupuesto”
- Slicer visual: A dropdown visual letting the user to select a value by budgeted ID, as informed in their corresponding column “clave”, with the help of the table “dim_clave”
 - id_presupuesto_descripcion: Calculated column created to concatenate the information of columns “id_presupuesto” and “descripcion” from table “dim_id_presupuesto”.

- Mostrar Presupuesto en Periodo: Inner filter that helps to retrieve the appropriate budgeted id considering the selection of period.
- Card visual: Informs the percentage corresponding of the top 10 specific budgets with higher expenses in the selected period.
 - % Monto Gastado por ID: Calculates the percentage corresponding to the selected budgeted IDs.
- Clustered bar chart: Informs the distribution between the top 10 specific budgets with higher expenses in the selected period.
 - Monto Gastado por ID.
 - % Monto Gastado por ID.
- **Presupuesto Especifico:** Section created to inform each specific budget assigned by the selected period.
- Table visual: Displays detailed information explaining the reason for use and the amount budgeted of each specific budget with their corresponding ID by the selected period.
 - mes: Calculated column to abbreviate the corresponding month and year assigned to each budget in table “fact_presupuesto”.
 - mes_continuo: Calculated column to sort in order of month the column “mes” in table “fact_presupuesto”.
 - Mostrar Presupuesto en Periodo: Used as inner filter.
- Detalle de Transacciones: Displays detailed budget-related transactions per row based on selections made on the report page.
 - Monto Gastado por ID.

What portion of my income has been generated through my own work over the years? How much corresponds to second-hand sales?

A report page called “**Reporte Ingresos Operativos**” was developed to analyze income generated exclusively from productive activities, such as personal work and second-hand sales. This page focuses on detailed transaction-level records to ensure accurate attribution of operational income over time. Non-operational inflows, such as “monedero” (cashback or similar third-party incentives), were intentionally excluded to prevent distortion of income metrics and to maintain analytical clarity in performance and trend evaluation.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total Ingreso Operativo:** Section created to inform the total incomes received over the years.
- **Card visual:** Display the total amount of incomes by the selected period.

- Total Ingreso Operativo: Calculate the total amount for the incomes registered, filtering out “monedero” as informed in “*tipo_ingreso*” in the table “*ds_ingresos*”.
- Line chart: Informs of the tendency of the incomes received by the selected period.
 - Total Ingreso Operativo.
- Gauge visual: Informs the corresponding amount for the selected type of income selected.
 - % Ingreso Operativo por Tipo: Calculates the percentage corresponding to a selected type of income.
- **Top 3 por tipo de ingresos:** Section created to inform the total amount spent by different types of income, by the selected period.
- Slicer visual: A dropdown visual created to select the different type of incomes obtained over the years, informed in the column “*tipo_ingreso*” in the table “*dim_ingresos*”.
- Clustered bar chart: Displays the total amount of the transaction made by type of income.
 - Total Ingreso Operativo.
 - % Ingreso Operativo por Tipo.
- **Top 5 ingresos por canal de ingreso:** Section created to inform the top 5 places where the user obtained money.
- Slicer visual: : A dropdown visual created to inform the different places where the user obtained money, as informed in the column “*canal_ingreso*” in the table “*dim_canal_ingreso*” by the selected period.
- Treemap: Shows the percentage corresponding to the magnitude of the top 5 places with higher incomes by the selected period.
 - Total Ingreso Operativo.
 - % Ingreso Operativo por canal de ingreso: Calculates the percentage corresponding to the magnitude of the places, by the selected period.
 - Magnitud Ingreso Operativo: Calculates the magnitude of incomes by place, by the selected period.
- Detalle de transacciones: Displays detailed income-related transactions filtered by selections made in the summary visual.
 - Total Ingreso Operativo.

What is the total amount spent on third parties and how are they distributed over the years? With whom have I incurred debts over the years? Do I have pending debts?

A report page **“Reporte Terceros”** was developed to analyze transactions involving third parties. This page presents detailed transactional data, enabling identification of the individuals or entities to whom the highest amounts were paid. It supports aggregation, ranking, and drill-through analysis, allowing the user to understand spending distribution and patterns by third party over time.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total gastado en terceros:** Section created to show how much was spent in third parties over the selected period.
- **Card visual:** Informs the total amount spent on third parties transactions over the selected period.
 - Total Terceros: Calculates the total amount spent by name.
 - % Monto Gastado en Terceros por Nombre: Calculates the percentage corresponding to each name by the selected period, as informed in the column **“nombre_abr”** in the table **“dim_nombre”**.
- Clustered column chart: Informs how much fluctuated the total amount spent on third parties by the selected period.
- **Top 10 gastado por persona:** Section created to inform the top 10 expenses made by third parties.
- Slicer visual: A dropdown visual letting the user to select a value by label, as informed in their corresponding column **“clave”**, with the help of the table **“dim_clave”**
 - nombre_con_abr: Calculated column created to concatenate the information obtained in **“nombre”** and **“nombre_abr”** in table **“dim_nombre”**
- Clustered column chart: Displays the total amount of the transaction made per name.
 - Total Terceros.
- **Top 5 gastado por clave:** Section created to inform the top 5 expenses made on third parties by the selected period of time.
- Slicer visual: A dropdown visual letting the user to select a value by level, as informed in their corresponding column **“clave”**, with the help of the table **“dim_nombre_clave”**
- Clustered bar chart: Informs the distribution between the top 5 labels with higher expenses in the selected period.
 - Total Terceros.

- % Monto Gastado en Terceros: Calculates the percentage corresponding to each label by the selected period.
- Historico gastado en deudas: Visual created to see the historic changes over time of all the transactions with a label "deuda". This visual works as Drill-Through to **"Reporte Deudas"**
 - Total Deuda: Calculates the total amount spent by debts.
- Detalle de transacciones: Displays detailed Third parties-related transactions filtered by selections made in the summary visual.
 - Total Terceros.

A report page **"Reporte Deudas"** was created. This page focuses on detailed records that compare amounts lent versus amounts repaid, including cases where settlement occurred through non-monetary compensation. The analysis is based on all the labels debt "deuda" informed in **"clave"** in the table **"fact_c_diaria"**

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total diferencia en deudas:** Section created to inform the total difference obtained for all the corresponding transaction by the selected period.
- **Card visual:** Informs the pending amount of debts in the selected period.
 - Total Deuda.
- Clustered column chart: Display the difference between debts over the selected period.
 - Total Deuda.
- Gauge visual: Informs the pending amount of debts in the selected period in percentages.
 - % Deuda: Calculate the percentage of the pending of debts by the selected period.
 - Deuda (<0): Calculates how much the user lent over the selected period of time.
 - Deuda (>0): Calculates how much the debtors paid in money over the selected period of time.
- **Top 5 deudores:** Section created to inform the top 5 debtors by the selected period of time.
- Slicer visual: A dropdown visual letting the user select a value by debtor by the selected period.
 - deudor_con_abr: Calculated column created to concatenate the information obtained in **"nombre_deudor"** and **"nombre_abr"** in table **"dim_deuda_nombre"**
- Clustered bar chart: Informs the distribution between the top 5 debtors with higher expenses in the selected period.

- Total Deuda.
 - % Monto Gastado en Deuda por Nombre: Calculates the percentage corresponding to each debtor by the selected period.
- **Top 10 gastado por ID deuda:** Section created to inform the top 10 expenses made by debts ID.
- Slicer visual: A dropdown visual letting the user to select a value by debt id.
 - Id_deuda_descripcion: Calculated column created to concatenate the information obtained in *"id_deuda"* from the table *"ds_deuda_id"* and *"descripcion"* from table *"dim_id_deuda"*
- Clustered column chart: Displays the total amount of the transaction made by debt ID.
 - Total Deuda.
 - % Monto Gastado en Id Deuda: Calculates the percentage corresponding to each debt ID by the selected period.
- Detalle de transacciones: Displays detailed debt-related transactions filtered by selections made in the summary visual.
 - Total Deuda.

Where have I traveled since the beginning of the Album Financiero record? How was my spending distributed across each trip and category? How much did I spend abroad, both in local and abroad local exchange?

A report page called **"Resumen Viajes"** was developed to provide a consolidated analytical overview of each trip. This page integrates transactional data with reference dimensions to present key travel attributes, including geographic location displayed on a map, the period during which each trip took place, and a representative image associated with each trip.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Filtros:** Section created to filter the page by the location of the trip and/or by the ids assigned to each trip.
- Slicer visual Filtro por ubicacion: A dropdown visual created to filter by continent, country, location (city or town).
 - ID Viaje Visible: Inner filter that filter out any location that wasn't used in the corresponding period.
- Slicer visual Filtro por grupo de viaje: A dropdown visual created to filter by grouped trips used between the selected period, were their corresponding value in column *"veces_usados"* in the table *"dim_viajes_grupo"* is greater than 1.

- grupo_viaje_con_descripcion: A calculated column that combines “grupo_viaje” and “viaje_principal” from the table “dim_viajes_grupo” into a single description.
 - ID Viaje Visible.
- Slicer visual Filtro por ID viaje: Slicer created to filter by each ID travel used between the selected period.
 - Id_viaje_con_descripcion: A calculated column that combines “id_viaje” and “descripcion” from the table “dim_id_viajes” into a single description.
 - ID Viaje Visible.
- **Mapa:** Section created to show the location of the places visited in each trip.
- Azure Maps: A map showing all locations where transactions were made during the selected trip.
 - ID Viaje Visible.
- **Informacion del viaje:** Section created to Inform the name of the place and the dates where the trip took place, selected in the slicer Filtro por ID Viaje.
 - Ubicacion Seleccionada: If the slicer is selecting trips that inform more than one place, a message appears informing them that multiple locations have been selected.
- **Fotografia:** Section created to show a representative picture saved in the folder “powerbi_viajes”, of the selected trip in the slicer Filtro por ID Viaje. If more than an ID viaje is selected, the first pictures’ visual will appear.
- **Historico gastado en viajes:** Visual created to see the historic changes over time of all the transactions made for a trip. This visual works as Drill-Through to “**Reporte Viajes**”
 - Total Viaje ID: Calculates the total amount spent by travel in the selected period.
- **Historico gastado en extranjero:** Visual created to see the historic changes over time of all the transactions made for a trip made abroad with a different currency exchange. This visual works as Drill-Through to “**Reporte Viajes Tipo Cambio**”
 - Total Gastado en el Extranjero: Calculates the total amount spent by travel made abroad in the selected period.

A report page called “**Reporte Viajes**” was developed to provide detailed analytical insight into the selected trip or set of trips. This page presents transaction-level information associated with each trip, enabling a deep-dive analysis of individual expenses and their characteristics.

It is important to note that aggregated totals on this page include all transactions linked to the selected trip, regardless of whether they fall within the date range displayed in the Ubicación Seleccionada visual. This design choice ensures that trip-

level financial analysis remains complete and accurate, even when certain expenses occurred outside the core travel dates.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total gastado en viaje:** Section created to inform the total spend over the selected period.
- Line chart: Displays total spendings made over the time between the selected period.
 - Total Viaje ID.
- **Card visual:** Displays total spendings made over the time between the selected period.
 - Total Viaje ID.
- **Gastado en moneda local vs. extranjera:** Section created to display how much was spent in the country of residence vs. spent in other countries with different currencies.
- 100% stacked bar chart: Displays how much corresponded to transactions made locally vs. the transactions made abroad in their local currency.
 - % Total Gastado en el país: Informs the corresponding percentage of the transactions made locally in the selected period.
 - % Total moneda extranjera: Informs the corresponding percentage of the transactions made abroad, in the selected period.
- Card Visual Total gastado en el país: Display the total amount spent locally between the selected period.
 - Total Gastado en el país: Calculates the corresponding amount made locally in the selected period.
- Card Visual Historico gastado en el extranjero: Display the total amount spent abroad, by the selected period. This visual works as Drill-Through to **“Reporte Viajes Tipo Cambio”**
 - Total Gastado en el Extranjero: Calculates the corresponding amount made abroad in the selected period.
- **Magnitud gastado en viajes por clave:** Section created to inform the expenses made by label by the selected period.
- Slicer visual: A dropdown visual created to select a label, as informed in **“clave”** in the column **“fact_c_diaria”**.
 - Viaje visible: Inner filter created to inform only of the transactions that are considered between the selections made between the page.
- Treemap: Displays the magnitude of each transaction made by label between the selected period of time, which higher the total made per label, bigger the section assigned will appear in the visual.

- % Magnitud Gastado Viaje ID: Informs in percentages the magnitude of each label appearing in the selected period.
 - Magnitud Gastado Viaje ID: calculates the magnitude of each label appearing in the selected period. ABS() is used to represent the magnitude of financial movements in the visual, ensuring that expenses stored as negative values are visualized proportionally by size without sign-related distortion.
 - Total Viaje ID.
- **Top 5 gastado por viaje:** Section created to inform the top 5 trips with higher expenses made by ID trip.
- Slicer visual: A dropdown visual created to select an ID trip available by the selected period.
 - Visible Viaje.
- Clustered column chart: Display the top ten higher expenses made by id trip, by the selected period.
 - Total Viaje ID.
 - % Monto Gastado en Viaje ID: Informs the corresponding percentage of the selected id trip.
- Detalle de transacciones: Displays detailed trip-related transactions filtered by selections made in the summary visual.
 - Total Viaje ID.

A report page called **“Reporte Viajes Tipo Cambio”** was developed to analyze transactions associated with the selected trip(s) that were executed in foreign currencies. This page focuses specifically on expenses incurred abroad, providing transaction-level detail to support accurate review of currency usage and conversion impact.

The page displays aggregated totals both in local currency and in the original foreign currency used for each transaction, enabling transparent comparison and clearer interpretation of international spending. This report follows the same analytical logic as the **“Reporte Viajes”** page, while narrowing its scope to foreign-currency transactions to support focused exchange-rate analysis and international travel insights.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Total gastado en el extranjero:** Section created to display the total spent abroad in their local currency.
- Slicer visual: A dropdown visual created to select the place where the travel took place (continent, country, state and city are available).

- ID Viaje Visible para Tipo de Cambio: Inner filter created to inform only of the transactions that are considered between the selections made between the page.
- Clustered column chart: Displays total spendings made over the time between the selected period.
 - Total Gastado en el Extranjero.
- **Card visual:** Displays total transactions made between the selected period.
 - Total Gastado en el Extranjero.
- **Magnitud gastado en viajes por clave:** Section created to inform the expenses made by label by the selected period.
- Slicer visual: A dropdown visual created to select a label, as informed in **"clave"** in the column **"fact_c_diaria"**.
 - ID Viaje Visible para Tipo de Cambio.
- Treemap: Displays the magnitude of each transaction made by label between the selected period of time, which higher the total made per label, bigger the section assigned will appear in the visual.
 - % Magnitud Gastado en Moneda Extranjera: Informs in percentages the magnitude of each label appearing in the selected period.
 - Magnitud Gastado en Moneda Extranjera: calculates the magnitude of each label appearing in the selected period. ABS() is used to represent the magnitude of financial movements in the visual, ensuring that expenses stored as negative values are visualized proportionally by size without sign-related distortion.
 - Total Gastado en el Extranjero.
- **Total gastado en moneda extranjera:** Section created to inform of the total amount spent by each foreign currency, in the selected period of time.
- Slicer visual: A dropdown visual created to select the foreign currency informed in **"moneda"** in the table **"dim_tipo_cambio_moneda"**, in the selected period.
 - ID Viaje Visible para Tipo de Cambio.
- Card visual mensaje: Displays a message informing when more than one currency is selected.
 - Leyenda Tipo Cambio: Creates a message if there are selected more than one currency.
- **Card visual Total gastado en moneda extranjera:** Displays the total amount spent on the corresponding foreign currency.
 - Total Gastado en la Moneda Extranjera: Calculates the total amount spent in the foreign currency, considering the columns **"moneda"** and

“*tipo_cambio*” from the table “*ds_viaje_tipo_cambio*”, in the selected period.

- Card visual Promedio tipo cambio: Displays the average exchange rate applicable in the selected period.
 - Promedio Tipo Cambio: Calculates the average exchange rate applicable in the selected period.
- **Top 5 gastado por viaje:** Section created to inform the top 5 trips with higher expenses made by ID trip abroad.
- Slicer visual: A dropdown visual created to select an ID trip available by the selected period.
 - ID Viaje Visible para Tipo de Cambio.
- Clustered column chart: Display the top ten higher expenses made by id trip, by the selected period.
 - Total Gastado en el Extranjero.
 - % Monto Gastado en Viaje ID en el Extranjero: Informs the corresponding percentage of the selected id trip.
 - Total Gastado en la Moneda Extranjera.
- Detalle de transacciones: Displays detailed trip-related transactions filtered by selections made in the summary visual.
 - Total Gastado en el Extranjero.
 - Total Gastado en la Moneda Extranjera.

What day is my final credit card payment due and how much should I pay?
Have I arranged the money to pay the credit debt on time?

A report page “**Reporte Creditos**” was created. This page focuses on detailed transactions informing the movements of credit made between period of credit dates informed by the credit line. After extensive testing and analysis, it was decided to maintain “*fecha_corte*” column from the table “*fact_creditos*” as the slicer field.

This decision was based on the fact that the credit cutoff date does not align with the first day of the month or any other standard calendar reference. Each credit period closes between the 6th and 8th day of the month, which makes it incompatible with the date structure of the general “*Calendario*” table. Using “*Calendario*” as a slicer introduced inconsistencies in filtering and prevented accurate alignment between credit periods and daily transactions.

Additionally, the possibility of transforming the curated dataset “*ds_creditos_periodo*” into a flat table was analyzed and discarded. Although this approach could simplify some relationships, it would break the logical integrity of the star schema by merging fact and dimension data. This structure is essential for

maintaining flexibility, scalability, and performance in Power BI. Flattening the table would reduce model efficiency, create redundancy, and limit the ability to perform dynamic cross-analysis between different datasets (credits, daily transactions, payment methods, etc.).

The month assigned to each period is the one informed by *“fecha_corte”* in the table *“fact_creditos”*

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Periodo de credito:** Section created to inform the period time informed by the credit line, between the selected period.
- Card visual: Displays the starting and ending date of the selected credit period. Also Informs the amount to be paid and the limit date to pay the total amount spent over the selected credit period.
 - **Total Creditos:** Calculates the total amount spent over the periods informed by the credit line.
- **Gasto por periodo de credito:** Section created to inform the distribution of total amounts divided by each credit month between the selected credit period.
- Slicer visual: Created to select the type of credit wanted to see in the chart visual the deposits and spends made (“deposito” or “gasto”) as informed in the column *“tipo_credito”* in the table *“dim_tipo_credito”*.
- Clustered column chart: Displays the total amount of the transaction made between each credit period, defined by *“fecha_corte”*.
 - Total Creditos.
- **Top 7 gastado a credito por clave:** Section created to inform the top 7 expenses made by credit by the selected period.
- Slicer visual: A dropdown visual letting the user to select a value by level, as informed in their corresponding column *“clave”*, with the help of the table *“dim_clave”*
- Clustered bar chart: Informs the distribution between the top 5 labels with higher expenses in the selected period.
 - Total Creditos.
 - % Monto Gastado por Credito: Calculates the percentage corresponding to each label by the selected credit period.
- Detalle de transacciones: Displays detailed credit-related transactions filtered by selections made in the summary visual.
 - Mostrar Tipo Credito: Inner filter that informs all transactions related to credit payments that occurred between the selected MIN and MAX dates.

Share

After addressing the key analytical questions through Power BI reports, supported by validated metrics and visuals, the next step was to prepare the solution for delivery and communication. This phase focuses on transforming the analytical outputs into clear, accessible, and presentation-ready artifacts that can be shared with a broader audience. This consists of the following core components:

1. Development of dashboard-style pages that consolidate the most relevant metrics derived from the analytical reports, allowing the user to quickly understand their current financial status and historical performance.
2. Enhancement of the overall visual design by applying consistent use of color, spacing, and layout principles. The objective is to improve readability, guide user attention, and ensure intuitive navigation across the solution.
3. Creation of a mobile version of the Dashboard.
4. Adaptation of both the master document and the Power BI solution into public versions, ensuring that sensitive information is excluded while preserving analytical value, structure, and storytelling.

Together, these elements represent the primary deliverables prepared for sharing. They ensure that the analytical work produced throughout the project is not only technically robust, but also clearly communicated, user-centered, and suitable for professional presentation and public dissemination.

Share – Dashboard and summary hub

During Phase I, in a single Dashboard page, the whole historical and actual finance information metrics and legends required by the user were initially addressed within a single dashboard page. However, following the analysis phase and the significant increase in the number of report pages developed during Phase II, it became clear that separating these goals would improve clarity, usability, and analytical focus.

As a result, the solution now includes two complementary summary pages. The first is a dashboard titled **“Situacion Financiera Actual”**, designed to answer the question “How am I financially right now?” by presenting a snapshot of the user’s current financial status through key metrics and indicators. The second is a summary hub titled **“Resumen Financiero”**, which provides a historical overview of financial performance and serves as a navigation entry point to the detailed analytical reports, helping to answer the question “Why is this happening?”

How am I financially right now? What balances do I hold across my payment methods and credit accounts? For how long can my current lifestyle be sustained based on my average monthly budget and available savings?

A dashboard page “**Situación Financiera Actual**” was developed to provide a concise overview of the user’s current financial status through key metrics and indicators. This single-page view enables the user to quickly assess:

- Outstanding credit balances
- Upcoming credit payment due dates
- An estimated timeframe for sustaining the current lifestyle based on available savings and average monthly budget
- Total amounts invested in active assets
- The percentage of the current month’s budget that has already been used

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Balance metodo de pago:** Section created to inform the payment methods available at the moment with current amounts (positive or negative).
- Table visual: Displays the current amount available in each payment method.
 - Balance Actual: Shows the current amount for each payment method that has a balance different from zero.
- **Estatus linea de credito:** Section created to inform the total credit limit and the available amount after subtracting pending payments charged to the credit.
- Card visual: Displays the current total credit limit and the available amount after subtracting pending payments charged to the credit.
 - Limite Credito: Informs the total credit limit informed in the column “*valor*” from the table “*fact_parametros*” tab.
 - Disponible Credito: Shows the remaining available credit.
 - Saldo Credito: Informs the amount spent on each credit payment method.
- **Acumulado en inversiones:** A section created to inform current values of each available investment.
- Table visual: Displays the current value of each available investment, informed as “Disponible” in the column “*tipo_inversion*” with the help of the table “*dim_tipo_inversion*”.
 - Balance Final en inversiones.

- **Resumen Estado de Pago del Credito:** A section created to inform how the status of payment of the current period is and the total amount accumulated for the next period.
- **Card visual:** Summarizes the payment status, total amount and due dates for the current credit period and the accumulated payment status corresponding to the next period date (if available).
 - **Resumen Pago Combinado:** Informs with messages the status of payment of the current credit period and the accumulated amount (if available) for the next period.
 - **Estado Pago Credito Mes Actual:** Combines both Monto Credito measures and gives format to the messages calculated.
 - **Monto Credito Mes Actual:** Calculate and informs with a message the amount to be paid for the current month, displays payment status and the limit date to be paid for the current month.
 - **ID Monto a Pagar en el Mes Actual:** Return the id assigned to the current credit period informed in the column *"id_creditos"* from the table *"fact_creditos"*.
 - **Monto Credito Siguiente Mes:** Calculate and informs with a message the accumulated amount to be paid in the next month.
 - **ID Monto a Pagar en el Siguiente Mes:** Return the id assigned for the coming ID from the current credit period informed in the column *"id_creditos"* from the table *"fact_creditos"*.
- **% utilizado del presupuesto:** Section created to display how much in percentage has been spent in the current month.
- **Gauge visual:** Displays the percentage of the budget that has been used in current month.
 - **% Presupuesto Mes Actual:** Calculates the percentage of the total budget used in the current month.
 - **Presupuesto Total Mes Actual:** Sum the measures Presupuesto Especifico Mes Actual and Presupuesto Regular Mes Actual.
 - **Presupuesto Especifico Mes Actual:** Calculate the sum of the specific budgets informed in *"tipo_razon_uso"* in the table *"dim_razon_uso"*.
 - **Presupuesto Regular Mes Actual:** Calculate the sum of the regular budgets informed in *"tipo_razon_uso"* in the table *"dim_razon_uso"*, in the current month.
 - **Monto Gastado Mes Actual:** Calculate the sum of the total transactions made in the current month.

- **Horizonte de Sostenibilidad y Fecha Limite:** Section created to display in messages how long can the user maintain living expenses for the future, considering budgets, money available in the payment methods (including the debt in credit) and available investments.
- Card Visual: Displays how long current investments and available debit can cover living expenses for the future.
 - Meses y Dias Sostenibilidad (con Fecha): Informs the the date values, formatted into months and days.
 - Meses de Sostenibilidad: Informs the estimated number of months (including fractions) that current funds can cover living expenses for the future.
 - Total Inversiones Disponibles: Total amount of investments currently available for spending.
 - Balance Actual.
 - Total Presupuesto Especifico: Calculate the sum of all the specific budgets planned for the future.
 - Total Presupuesto Regular: Calculate the monthly sum of the regular budgets planned.
- Ir a Resumen de Viaje: Navigation page created to be sent to the report page **“Resumen de Viaje”**.
- Ir a Resumen Financiero: Navigation page created to be sent to the summary hub page **“Resumen Financiero”**.
- Ultima actualizacion: Display the last time the power BI tables were refreshed manually by the user.
 - Ultima Actualizacion: Calculates the last table the Refresh option was succesfully applied by the user in Power BI, as informed in the column **“refresh_info”** in the specially created for this calculation the table **“refresh_info”**.

How have my personal financial transactions evolved over the years across the main areas of activity? What is the overall balance between income, withdrawals, third-party spending, and travel-related expenses over time?

The summary hub page **“Resumen Financiero”** was developed to present a consolidated view of historical personal financial movements recorded over time. The page name was retained from the original dashboard developed in Phase I, as it continues to serve the same purpose: providing a high-level financial overview.

This page integrates the main analytical themes derived from the master dataset, including balances between incomes and withdrawals, spending associated with third parties, and expenses related to trips.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Balance de ingresos y egresos:** Section created to inform the comparative between the total incomes and withdrawals occurred by the selected period.
- **Line Chart:** Displays the comparative of total incomes and withdrawals (with absolute values so can appear in the same quarter) by the selected period.
 - **Total ingreso:** Calculates the total incomes that generated gains in the master document, excluding the transactions where *"razon_uso"* = DINERO and *"clave"* = "transaccion".
 - **Total gastado:** Calculates the absolute amounts that generated losses in the master document, excluding the transactions where *"razon_uso"* = DINERO and *"clave"* = "transaccion".
- **Card visual:** Inform the total amounts that generated gains, losses and the total net generated by the selected period.
 - Total ingreso.
 - Total gastado.
 - Total Neto.
- **Historico de ingresos y egresos:** Section created to inform the total amounts applied in three main concepts defined for this project: incomes obtained, investments and the budgeted transactions.
- **Clustered bar chart:** Displays the total amounts applied in three main concepts defined for this project: incomes obtained, investments and the budgeted transactions, by the selected period.
 - Total Inversiones Monedero.
 - Monto Gastado.
 - Total Ingreso Operativo.
- **Card visual:** Inform the totals amount of the three main defined concepts in the project.
 - Total Inversiones Monedero.
 - Monto Gastado.
 - Total Ingreso Operativo.
- **Historico gastado en terceros:** Section created to inform the total amounts used under the concept of third parties, by the selected period.
- **Stacked area chart:** Displays the total amounts used under the concept of third parties over the selected period.
 - Total Terceros.
 - Total Deuda.
- **Card Visual:** Informs the total amounts used under the concept of third parties, by the selected period.
 - Total Terceros.

- Total Deuda.
- **Historico gastado en viajes:** Section created to inform the total amount spents in trips, by the selected period.
- **Clustered column chart:** Displays the total spent under the concept of trips over the selected period.
 - Total Viaje ID.
- **Card visual:** Informs of the total amount spent per trip, by the selected period.
 - Total Viaje ID.

Share – Visual Standards and Interface Elements

Visual and metric design decisions across the Power BI solution were intentionally standardized to ensure consistency, readability, and a clear distinction between analytical content and interactive elements.

- All numeric and text values displayed in cards and key indicators use white (#FFFFFF) to maximize contrast against the background templates. An exception was applied to the total values displayed in the table visual “Detalle de Transacción,” where the same light yellow used in the “monto” column (#FDD472) was retained to preserve visual continuity and highlight aggregated values. Card visuals use a text size of 18, while table values use a text size of 10; both are displayed using the Arial font.
- Chart visuals primarily use a defined color palette consisting of light blue (#70B0E0), yellow (#FCB714), and blue (#2878BD). Visuals that require a broader range of categories apply the colors provided by the Innovate theme selected in the Power BI View settings, ensuring consistency with the overall report styling.
- Titles for visuals and sections are displayed in gray (#808080), using the Arial font and primarily a text size of 16. Card labels are displayed in light gray (#D2D0CE), also using Arial, with a primary text size of 10, allowing labels to remain visible without competing with key metric values.
- Dropdown slicers use a dark gray color (#3A3A3A) to clearly distinguish filtering controls from analytical visuals while maintaining alignment with the overall color scheme.

Navigation buttons

The Power BI solution incorporates navigation buttons positioned in the upper-right area of each report page to support intuitive movement across the report and improve overall user interaction. All buttons used were selected from the standard options available in Power BI under the Insert > Buttons menu, ensuring consistent behavior and native compatibility.

- A left arrow button was implemented to allow the user to return to the main dashboard page, **“Situación Financiera Actual”**.
- A bookmark button was used to provide direct navigation back to the summary hub page, **“Resumen Financiero”**.
- A reset button was configured to clear all slicer selections applied on the current page, allowing the user to quickly return to the default view.

On the **“Situación Financiera Actual”** page, two additional navigation buttons were created to enable direct access to the **“Resumen de Viajes”** and **“Resumen Financiero”** pages. These shortcuts were added to reduce navigation steps and improve access to the most frequently referenced summary views.

Templates

Visual templates were created for Power BI to be used as canvas backgrounds across the report pages. These templates were designed to ensure visual consistency, improve readability, and provide a standardized layout framework for all visuals and sections within the solution. The next colors were used between the templates:

- The primary background color used in the templates is blue (#03222F), which serves as the base canvas for all Power BI pages.
- A darker blue tone (#042433) was applied to define and separate the main sections within each page, helping to visually structure the content.
- A red color (#260F04) was assigned to visuals used for drill-through actions and page navigation, allowing these interactive elements to be easily identified by the user.
- A green color (#061808) was consistently applied to all date-related filters to distinguish filtering controls from analytical visuals.

The color codes were defined and applied within templates created in PowerPoint. Due to variations in layout across report pages, a dedicated slide was created for each page to reflect its specific section distribution. These slides were compiled and stored in a document titled **“plantillas”**, which serves as the reference source for all visual layouts used in the Power BI solution.

Once finalized, the templates were exported and saved as SVG files, enabling their use as background images within Power BI while preserving layout accuracy and visual quality.

Share – Mobile version

As in Phase I, the mobile version was designed to address the user's need to quickly access the most up-to-date snapshots of the current financial status. For this reason, the dashboard page **"Situacion Financiera Actual"** was duplicated and reformatted for mobile consumption within the Power BI solution. This new page was renamed **"Version Movil"**.

The mobile version is focused on presenting the same visuals and metrics available in the desktop version of the dashboard, but it was decided to add a second section to function as a search area. This section allows interaction through date selection and responds to one of the original user requirements identified at the beginning of Phase II.

The design of the mobile version was intentionally kept simple, prioritizing clarity, readability, and ease of interaction on smaller screens. While functional completeness was the primary goal at this stage, further visual and usability enhancements are planned for future iterations of the project.

The following sections and visuals were developed, and the DAX measures used in each of them (if applicable) will be explained:

- **Buscador de descripcion:** Section created to be able to search for a key word or complete description and been able to see the corresponding ID and amount per transaction, by the selected period.
- Slicer visual: A between slicer created to select the year between the existing dates informed in the master document.
- Slicer visual: A Tile visual showing each month in different tyles, so from the selected year can be selected a specific moth.
- Search bar: Icon created to be able for search with a key word or concept a transaction the user want to look out from the column **"descripcion"** from the table **"fact_c_diaria"**, by the selected period.
- Table visual: Informs the description, amounts the period and ID used in the master document, by the selected period.
 - Total Cuentas Diarias.

Share – Master Excel documents and Powe BI solution

The design of the original “**album_financiero**” spreadsheet was redistributed to leave a main master document and separate spreadsheets with dimension/reference tables:

- “**master_album_finaciero**” is now considered as the main Excel document and have the next spreadsheets considered as fact tables:
 - “**c_diaria**”: represents a transaction, serving as the project’s fact table.
 - “**presupuesto**”: contains monthly and annual budgets established by the user to maintain control of personal finances.
 - “**inversiones**”: contains details the investments that have existed in the life of the “Album Financiero” spreadsheet.
 - “**credito**”: contains both settled and outstanding credit debts informed by credit periods defined by the credit line.
 - “**parametros**”: informs the total available amount for each managed credit account.
 - The “**definiciones**” tab consolidates definitions and recommendations for the columns “*razon_uso*,” “*clave*,” and “*metodo_pago*.” For that reason, this spreadsheet was left in the master document.
- “**tabla_geografia**” store complete geographical information for each place visited and recorder in the “**tabla_viajes**” document.
- “**tabla_id_deuda**” store the complete information of each “*id_deuda*” used in the master document.
- “**tabla_id_presupuesto**” store the complete information of each “*id_presupuesto*” used in the master document.
- “**tabla_viajes**” provides a historical log of trips taken over the years, including travel dates and brief descriptions or reasons for each trip.
- The helping columns of the spreadsheet “**c_diaria**”, “*id_presupuesto*”, “*m.extranjera*”, were deleted as they were replaced in the creation of the curated datasets process.
- Both the Excel and Power BI files were moved to their final file location. Both were renamed into “**master_album_financiero**”.
- The documents created for the project were moved to their final file location:

- “csv_output”: where the CSV documents are saved after been generated with the “actualizar_album_financiero” document.
 - “actualizar_album_financiero”: Exe document created to generate the whole procedure of pipeline orchestration to generate the final curated tables to be used in Power BI.
 - “powerbi_viajes”: where all the images used for the “Resumen de Viajes” report page were stored.
- The Power BI solution was already published in Power BI Service and is available to be seen via online and in a smartphone.
 - The option to automate the refresh and Publish option of Power BI cannot be done without a Business License. As we are working with a personal license and it was decided not to proceed with the change of license, this procedure will be done manually.

Share – Public Master Excel documents and Powe BI solution

- Copies of the original Power BI (pbix) and Excel (xls) files were created and they maintained their original name with the addition of “_publico” for publication purposes.
- All monetary values were modified to protect personal information. Generally speaking, the following logic was applied:
 - The sign of each amount (positive or negative) was preserved, to maintain consistency with the original transaction type.
 - All values were systematically scaled and rounded, ensuring the analytical logic and proportions remain intact while obfuscating actual figures.
 - Minor manual adjustments were made to remove small non-representative amounts introduced during transformation that affected the Power BI metrics. With those minor changes, the overall logic or results weren’t affected by the changes.
- Text columns were also adjusted in each table :
 - In the documents “tabla_viajes_publico” and “tabla_geografia_publico”: Travel locations and descriptions were randomly changed using formulas and Google Maps.
 - In tables “definiciones”, “c_diaria” and “parametros” in the “master_album_financiero_publico” : The “metodo_pago” categories were renamed as “entidad” showing whether the method was debit, credit or cash.
 - In spreadsheet “c_diaria” : The column with texts “descripcion” and “lugar_de_uso” were renamed with the help of the columns “clave” and

- “razon_uso”. The column “monto” was adapted with formulas to inform an amount following the same logic as the original so the metrics wouldn’t suffer big changes in their logic.
 - In pages “inversiones” and “c_diaria” : The “tipo_inversion” column was renamed to generic labels like inversion_1, inversion_2, etc. These changes also affected the associated descriptions.
 - In page “c_diaria” : The “descripcion” column was adapted in transactions the main key words (like “Ahorro” or “loc” for investments movements) and/or signs (“.” for ID budgets and “:” for third parties names) that directly affected the spreadsheets “inversiones” and “presupuesto” and the table “tabla_id_presupuesto”.
 - In page “presupuesto”: The “descripcion” column was renamed based on its “razon_uso”.
- A folder “powerbi_viajes” was created with random images, so the power bi solution could work with the folder in the visualization used in the report page “Resumen de Viajes”.
- The document “relaciones_tablas_powerbi” was created to inform to have a well-organized spreadsheet to easily find the relationships made in the Model View in the public Power BI solution.

One of the principal objectives of the project was to automate the generation of curated datasets by transforming master Excel documents into CSV files, importing them into PostgreSQL, and creating structured tables to be consumed by Power BI. This objective was fully achieved as expected. However, because this process depends on credentials, local folders, and private file paths, the decision was made not to include this automation pipeline in the public version of the project.

For the public version, the curated data was instead uploaded directly into Power BI using CSV files. These files preserve the same structure, formats, and column definitions described throughout this report. As a result, the Power BI solution did not require any changes to the existing data model, table relationships, or DAX measures.

To organize the generated datasets, the folder “datasets_generados” was created. This folder contains the curated datasets used throughout the project and is structured as follows:

- ds_tables and dim_tables: Contains the final version of the tables used to feed the Power BI solution. These tables were designed to support relationships following a Star Schema approach and to provide only the data required for the correct functioning of visuals and DAX measures.
- v_ds_tables: Contains extended versions of the ds_tables, created to provide additional contextual information. These tables are intended to support data

analysis, validation, and troubleshooting, making it easier to understand the underlying data and resolve queries. They are designed for analyst use and are not intended for final-user consumption.

Share – SQL, Python and DAX Implementation Scripts

From the outset of the project, one of the primary objectives was to strengthen practical coding skills commonly required in data analytics roles, with a focus on optimizing workflows and producing reliable, scalable metrics. To support this objective, the solution intentionally integrates SQL, Python, and DAX across the data transformation, automation, and analytical modeling layers.

The creation of curated datasets was primarily implemented using PostgreSQL through DBeaver. All SQL scripts used to generate the datasets consumed by the Power BI solution are available in the folder “[SQL Scripts](#)”. These scripts are organized into two main groups, identified by a numeric prefix in the file name:

- Scripts starting with 1. correspond to the creation of the PostgreSQL database, the import of the master source documents, and the generation of the base tables used by the Power BI solution. This group also includes the scripts used to automate table population and data updates.
- Scripts starting with 2. correspond to the generation of curated datasets and dimension tables required for the Power BI data model to function as expected.

Python was used to automate the generation of documents and to manage the update process within PostgreSQL. This automation was implemented through an executable file created from Python scripts. As mentioned previously, the executable is not available for direct use in the public version of the project. However, the scripts used to generate it are included in the folder “[Python Scripts](#)”, where the main script, “[actualizar_album_financiero](#)”, is available and ready to be used to recreate the executable if required. Inclusive the scripts used to create the folder “[powerbi_viajes](#)” and the document “[images_url_publico](#)” are available to be implemented.

Together, these scripts and formulas document the technical foundation of the project and provide transparency into the processes used to transform raw data into curated datasets and analytical insights.

The **DAX measures** used throughout both phases of the project are fully documented and separated by phase. Measure definitions are stored in the document “[dax_measures](#)”, while extended contextual documentation is available in the Excel file “[dax_mediciones_finales](#)”. Within this file:

- The worksheet “dax_phase_I” documents all measures created during Phase I, preserving the same structure and organization used in the original “dax_mediciones_y_visuales” document.
- The worksheet “dax_phase_II” focuses on Phase II and documents all DAX measures using the following structure:
 - *dax_nombre*: Informs the name of the corresponding measure.
 - *dax_medicion*: Informs the measure used in the DAX measure.
 - *nombre_reporte*: Informs the name of the corresponding report page where the measure was used.
 - *seccion*: Informs to which section of the report was used the measure.
 - *nombre_visual*: Informs in which visual was used the measure.
 - *uso_medicion*: Informs if the measure was used directly in the visual, or as an inner filter, or if it was used as a calculated column.
 - *veces_usado*: Informs how many times the measure were used over the Power BI solution.

The total number of DAX measures increased from 53 measures in Phase I to 97 measures in Phase II. This increase is primarily explained by two factors:

- The number of report pages expanded significantly, from **8 pages** (1 dashboard, 4 report pages, and 3 tooltips) to **14 pages** (1 dashboard, 1 summary hub, and 12 report pages), reflecting the broader analytical scope and the need for clearer separation of key questions across dedicated views.
- The DAX strategy evolved toward improved readability and maintainability. In Phase I, many measures were highly complex and self-contained, as all logic needed to be executed directly within Power BI without support from external tools. In Phase II, after offloading heavy transformation logic to PostgreSQL, DAX measures were deliberately modularized. This separation allows each measure to perform a specific, well-defined task, making the overall calculation flow easier to understand, review, and optimize.

This layered approach ensures that the analytical solution remains transparent, maintainable, and aligned with professional BI and analytics engineering practices.

Act - Recommendations and Next Steps

Although the objectives defined at the beginning of the project and the recommendations identified at the end of Phase I were successfully addressed during Phase II, additional opportunities for improvement were identified during implementation and post-analysis. These recommendations are proposed as future enhancements to further increase analytical depth, usability, and long-term scalability of the Master Album Financiero solution.

Data Modeling & Analytical Depth

- Evaluate the creation of a **“tabla_id_ventas”** to explicitly track purchase and sale events over time. This table would improve traceability of commercial activity and strengthen analysis in the **“Reporte Ingresos Operativos”** particularly for transactions classified as “Actividad Comercial” in the **“tipo_ingreso”** field of the **“dim_ingresos”** table.
- Refine the handling of the concept “varios” in debt analysis, replacing it as an independent entity and instead assigning transactions directly to the third party responsible for the debt. This adjustment would improve interpretability and accuracy in the **“Reporte Deudas”** specifically within the **“nombre_deudor”** column of the **“dim_deuda_nombre”** table.
- Enhance the **“tabla_id_viajes”** table by separating transactions related to origin-city expenses (such as transportation or preparation costs) from destination-level travel expenses. Currently, these transactions are attributed to the first destination, which can distort country-level analysis—particularly in the **“Reporte Viajes en el Extranjero”** where multi-country spending may be aggregated incorrectly.

Reporting & Visualization Improvements

- Iterate on the visual templates used across the Power BI solution to achieve a more refined, professional, and visually balanced design. These improvements would be guided by user feedback collected through continued usage of reports.
- Enhance the mobile version of the Power BI solution, improving layout, spacing, and visual hierarchy to provide a more polished and user-friendly experience while maintaining analytical clarity.

Data Quality & Semantic Clarity

- Continue improving the standardization and clarity of the “*descripcion*” columns in the Master Documents. Further refinement would improve readability, reduce ambiguity, and enhance slicing, filtering, and drill-through behavior within Power BI.

Advanced Capabilities & Innovation

- Explore the direct integration of AI-assisted features within both the master documents and the Power BI solution. This includes leveraging native AI tools available in the platforms to support automated insights, anomaly detection, assisted querying, and future analytical enhancements across the project ecosystem.

Conclusions & Reflections

Over the past six months of work on Phase II of this personal capstone project, I have confirmed that the role of a data analyst—particularly within a Business Intelligence context—is both broad and deeply rewarding. The project required not only technical execution, but continuous evaluation of data quality, modeling decisions, analytical relevance, and communication effectiveness. As happened during Phase I, I frequently revisited personal notes, reviewed prior coursework, rewatched instructional material, and explored different approaches to storytelling and data visualization. This iterative learning process directly influenced the quality of the decisions made and strengthened the overall solution.

Looking back at the full lifecycle of this project, from its initial conception approximately nine months ago to the much longer period of personal financial tracking that began in May 2015, this work became more than technical exercise. Through the process of structuring, modeling, and analyzing a decade of personal financial data, I was able to clearly identify my professional focus and motivation. I discovered a strong interest in Business Intelligence as a discipline—one centered on transforming complex, long-term datasets into meaningful metrics, intuitive visualizations, and reliable insights that directly address stakeholder needs.

This project demonstrated how thoughtful data modeling, careful metric design, and intentional visualization choices can extract maximum value from existing data sources. More importantly, it reinforced the idea that strong BI solutions are not defined solely by technical complexity, but by their ability to balance analytical depth, usability, performance, and clarity. The evolution of the Album Financiero project reflects this philosophy, transitioning from a manually maintained tracking system into a scalable, automated, and insight-driven BI environment.

Ultimately, this capstone represents both a technical milestone and a personal turning point. It consolidates years of data, months of structured learning, and a clear decision to pursue a BI-oriented professional path—one focused on delivering accurate, well-modeled, and actionable insights that support informed decision-making over time.

On the Use of Generative AI

As in Phase I, generative AI played a supportive role throughout this project, primarily as a learning and problem-solving aid. Its use was focused on assisting with the development and refinement of complex SQL queries, Python scripts, Power Query transformations in Excel, Power BI functions, DAX measures, and data modeling logic.

Used as a collaborative assistant—particularly through ChatGPT—generative AI helped accelerate troubleshooting, clarify technical concepts, and explore alternative implementation approaches. This allowed more time to be dedicated to analytical design, data modeling decisions, and the overall coherence of the Business Intelligence solution.

All architectural decisions, analytical interpretations, and final implementations were defined and executed by the analyst. The use of generative AI did not replace domain understanding or judgment, but rather enhanced productivity, supported continuous learning, and contributed to the depth, quality, and completeness of the final solution.

Reflections on tool selections

In Phase II, the project expanded beyond Excel-centric processing into a more robust Business Intelligence architecture. SQL became the core technology enabling this transition, as it was used to design, generate, and maintain curated datasets that serve as the primary data source for the Power BI solution. By connecting Power BI directly to PostgreSQL through DBeaver, the analytical model was decoupled from the Excel files and aligned with scalable, enterprise-style BI practices.

Although the current dataset contains thousands of records, this architectural decision was made with scalability in mind. In business environments where datasets commonly grow to hundreds of thousands or millions of rows, relying on Excel-based transformations would significantly increase refresh times and resource consumption. Preprocessing data in SQL allows logic to be materialized upstream, reducing computational load in Power BI, improving performance, and ensuring the solution remains maintainable as data volume and complexity grow.

Python was primarily used to automate the orchestration of the data pipeline. Its role focused on eliminating manual intervention in the process of exporting Excel files, loading them into PostgreSQL, executing transformation logic, and generating analysis-ready datasets. By packaging this workflow into an executable file, the update process was reduced to a single action, improving reliability, reducing the risk of human error, and reinforcing reproducibility across refresh cycles.

DAX Studio was used as a complementary tool to Power BI to efficiently extract and document table relationships and DAX measures. This facilitated faster inspection of the semantic model, ensured adherence to the Star Schema design, and supported the creation of structured documentation for all measures used throughout the project. These outputs form part of the project's technical artifacts and improve transparency, maintainability, and long-term usability of the solution.

Finally, generative AI tools—particularly ChatGPT—were used as a technical support resource when working with complex SQL logic, Python automation scripts,

Power Query transformations, and DAX expressions. Their use accelerated problem-solving and helped validate implementation approaches, while all architectural decisions, logic design, and final implementations remained fully defined and executed by the analyst.

Development Steps & learnings

As a final personal touch, I divided this project into four steps as in each one I had to rethink again what I was working on and what I really wanted to see in Excel and the Power BI Solution.

1. Goal Definition and Planning: identifying key questions and requirements. My main objective became clear: to professionalize my Excel-based tracking system. This meant transforming it into a more intuitive, visually organized structure, while transferring all calculations, formulas, macros, and pivot tables from Excel to Power BI.
2. The extract, transform, and load (ETL): I cleaned and restructured the data source to make it ready for analysis. This step also included formulating five main inquiries that would guide the visual design and structure of the Power BI solution.
3. Power BI Development (Analysis): This is where the ideas took form. Through DAX formulas, proper table relationships, and intentional visual design, the dynamic behavior I envisioned came to life—answering the five key questions and unlocking new insights from my historical financial data.
4. Final Edits and Public Version: In the final phase, I focused on reviewing the Power BI solution from a user perspective rather than a developer one. This allowed me to make final improvements—adjusting visuals and measures, titles, and layout—to ensure that the story being told was not only accurate but also engaging and easy to follow. This phase also included the creation of a public-facing version of the project called “album_financiero_publico”.

Having a clear and well-defined goal was essential not only for maintaining direction, but also for adapting effectively when changes to the original work plan were necessary. Throughout the project, new ideas and challenges required flexibility and rethinking of many steps. However, the established objective served as a solid reference point, helping me stay on track despite those changes. This focus proved invaluable in keeping the work structured, intentional, and ultimately successful.

Appendix and comments of the author

1. The following logic was applied to the style and formatting of the names related to the project, to create a more visual distinction of their origin:
 - **Excel document:** Refers to an Excel document.
 - **Spreadsheet:** Refers to a sheet in the Excel document.
 - *Column:* Refers to a column in the Excel document.
 - CATEGORY: Refers to a category within the *razon_uso* column.
 - Label: Refers to a key within the clave column.
 - **Folder, documents (no excel) or Exe icon:** Refer to a folder or Exe icon created.
 - **Page:** Refers to a page in the Power BI document.
 - **Table:** Refers to a table in the Power BI document.
 - **Column:** Refers to a column in a table within the Power BI document.
 - **Section:** Refers to a section created in a page in Power BI document, where different visuals were used to explain the reason of the section.
 - **Drill-through or Page Navigation:** Refers to a visual that has the measure working as a drill-through or has a page navigation button in the Power BI document.
 - **Total per transaction:** Refers to a card visual working as the total amount informed in the corresponding “Detalle de transacciones” in the page.
2. To maintain consistency and enable future automation with different analysis tools, the columns, categories and labels were standardized using snake_case conventions, except for “razon_uso” categories.
3. All sheets, tables, columns, categories, labels, visualizations, SQL statements Python codes and DAX measures names were written without accent marks. This decision was made to simplify their use and to ensure compatibility with future analytics tools.
4. As this is a personal project, sheets, tables, columns, categories, labels, visualizations, and DAX measures names were not translated into English. This document represents the original version, which was developed in English from the start, in collaboration with ChatGPT, as part of an exercise to practice the professional terminology used by Business Intelligence analyst.
5. A [Github repository](#) was created to archive and organize the whole documentation created for Phase II of the project.
6. The whole public Master Documents excel tables, Power BI Solution and the curated datasets feeding the Power BI Solution, are available in the folder

- “data” in the Github repository corresponding to the Phase II of the general Album Financiero project.
7. The “master_album_financiero – Full Report” document is available in the folder “docs” in the Github repository corresponding to the Phase II of the general Album Financiero project.
 8. If is required a deeper analysis of the relationships of the tables used in the Power BI solution, the document “relaciones_tablas_powerbi” is located in the folder “data” in the Github repository corresponding to the Phase II of the general Album Financiero project.
 9. GIF versions for each report page are available in the folder “images”, in the Github repository corresponding to the Phase II of the general Album Financiero project.
 10. If is required, the templates used for the project are located in the “Templates” folder inside the “images” folder in the Github repository corresponding to the Phase II of the general Album Financiero project.
 11. If a deep analysis of the Dax measures is required, these can be found in the folder “SQL Scripts” in the Github repository corresponding to the Phase II of the general Album Financiero project. Also, in the same folder can also be found the “tablas_generadas” that serves as an organization table for the curated dataset created.
 12. If a deep analysis of the Python scripts is required, these can be found in the folder “Python Scripts” in the Github repository corresponding to the Phase II of the general Album Financiero project.
 13. If a deep analysis of the Dax measures is required, these can be found in the folder “DAX Measures” in the Github repository corresponding to the Phase II of the general Album Financiero project.
 14. A [Google Sites portfolio](#) was created to inform a brief summarize of the project with images and comments of the author, in its corresponding section.
 15. A walkthrough video of the project was created and ready to be viewed is already in the corresponding section of Phase II in the Google Sites portfolio, showcasing the evolution of the data source and dashboard: before, during, and after the capstone process. This will help communicate project goals, methodology, and outcomes to external reviewers or future collaborators.